Customer touchpoints and mobility service value

A proposed service model connects consented intent to reachable supply, additional activity and feasible travel choices.

01 / PRODUCT OUTCOME & SCOPE

PRODUCT HYPOTHESIS

Customer intent and shared transport capacity can improve service outcomes through five testable mechanisms.

CONSENTED SIGNALS Intended arrival, booking lead time, free-time interval and optional preferences. Purpose, expiry and withdrawal remain explicit.

Resident / visitor

Share plans, free time and preferences with consent. Retain control of timing, price and accessibility needs.

Mobility operator

Position reachable vehicles, protect committed trips and own exceptions within finite operating capacity.

Service provider

Return activity slots, prices and service capacity. Confirm reservations and observed use.

City partner

Set ODD, access and coverage conditions. Review outcomes for underserved places and users.

MaaS product layer

C01 identity / API → C02 booking & consent → C03 feasibility / reservation → C04 dispatch & status → C06 support / cases → C07 usage / settlement | C05 energy capability constrains feasible supply

DECISION INPUTS Intent, reachable supply, provider slots, hub / energy limits and accepted terms

END-TO-END SERVICE LOOP

01	02	03	04	05
Intent	Feasible offer	Confirmed plan	Trip + support	Verified outcome

FIVE VALUE MECHANISMS / INFORMATION AND SHARED CAPACITY

01 Advance positioning

INPUT Received plan + lead time.
ACTION Position reachable supply before pickup.
TEST Wait, empty km and unmet demand.
LIMIT Cancellation, forecast error and energy.

02 Additional demand

INPUT Free time + preference + spare capacity.
ACTION Offer a feasible activity and trip.
TEST Additional use and net contribution.
LIMIT Substitution and induced congestion.

03 Demand smoothing

INPUT Flexible window + alternative total price.
ACTION Move an accepted trip in time.
TEST Peak load and missed pickups.
LIMIT Refusal and discount cost.

04 SHARED RIDES / 05 PASSENGER–FREIGHT SHARING

Match compatible passenger journeys within seat, access and detour limits. Test passenger output, discounts and net contribution. 05: Insert eligible cargo within payload, space and handling limits. Include freight revenue, added dwell and displaced passenger trips. Detailed operating conditions: P29–P30.

SERVICE CONTRACT C02 keeps each passenger / shipment intent. C03 checks route and segment capacity. C04 records the shared mission. C06 owns passenger recovery and cargo exceptions. C07 reconciles revenue and shared costs. C05 constrains energy.
VALUE TEST Protect consent and accepted service limits. Report additional demand, shifted demand and shared-route efficiency separately. Include rejection, cancellation, handling and recovery. Capacity / continuity: P25–P28. Sharing rules: P29–P30.

End-to-end journey across user, operation and platform

Passenger bookings, activity reservations and shipments retain distinct identities through shared routes, revised offers and verified outcomes.

02 / SERVICE BLUEPRINT



STAGE EVIDENCE **Plan: consent + receipt time** **Request: eligible offer + capacity** **Confirm: accepted terms + committed holds** **Travel: observed pickup + owned exception** **Complete: trip / activity actuals + cost**

EXCEPTION LOOP

A revised plan releases obsolete holds only after reconciliation. Cancellation or capacity loss updates positioning, ETA and the recovery owner.

TRACEABILITY CONTRACT

intentId + passenger / shipmentId + bookingRev + routeRev + missionId

Record offers, accepted terms, original / revised times, route stops and segment occupancy. Link every passenger and shipment to observed boarding, alighting or delivery. Preserve source / receipt time, consent and custody evidence. Shared mission costs remain unique.

ACCEPTANCE INVARIANTS

Positioning uses information already received. Commit all required holds before confirmation. Check seats, accessible space, payload and cargo volume on every segment. Recompute existing passenger limits after insertion. Time shifts preserve request identity. Suggestions need acceptance. Unknown effects require reconciliation. Completion uses passenger / cargo evidence separately. Each exception retains an owner and deadline.

MVP boundary and product prioritization

A complete service slice provides the baseline. Five controlled mechanisms test information value and shared-capacity economics.

03 / MVP & PRIORITIZATION

Priority rule

Rank by measured service benefit, net contribution, feasibility and learning value.

Common fleet / hubs / demand conditions support comparison. Capacity and service guardrails remain binding.

DEPENDENCY ORDER **Reliable request / outcome records precede effect measurement. Finite capacity and recovery precede offer expansion.**

PILOT MVP

One selected service slice

01	Capture intent, needs, consent and idempotent request
02	Return feasible options, binding reasons and expiry
03	Commit booking + resource bundle with revision checks
04	Track state, freshness, support case and recovery owner
05	Reconcile actual trip, service use and passenger outcome

Exit only when normal + recovery paths pass acceptance; operator can resolve exceptions; event and cost records reconcile.
Required evidence: valid, rejected, cancelled and recovered journeys. Reconcile every request to one final disposition and every cost to one owner.

ITERATE

Five controlled mechanisms

01	Use received plans for feasible advance positioning
02	Offer activities within spare time and capacity
03	Shift accepted trips with clear timing and price
04	Pool compatible riders within each detour limit
05	Insert eligible cargo within payload and time limits

Advance when gains survive service guardrails, realistic acceptance, full costs and supply stress.
Retain independent controls for all five effects. Test late signals, substitution, peak overload, no-shows, cargo dwell and failed handovers.

EXTEND PLATFORM

Repeatable operating model

01	Versioned adapters and purpose-limited consent
02	Energy, hub throughput and staffing in feasibility
03	Guarded support and owned passenger recovery
04	Calibrate by zone, time and user segment
05	Reusable configuration, evidence and rollback

Expand after second-site tests preserve service quality and net contribution with local demand, capacity and operating costs.
Replication tests: different OD pattern, provider capacity, support load and energy limits. Retain tenant isolation and configuration / policy versions.

MVP CONTROL

MVP entry criteria: selected geography / ODD / channel / segment; named service and incident owners; provider, fleet and site capacity confirmed; legal, privacy, accessibility and partner constraints reviewed; C01–C07 data and authority boundaries agreed. Pilot exit: normal + cancellation + disruption / recovery traces pass; queue and support coverage are staffed; KPI baseline, missing-data behavior and cost ledger reconcile; rollback and customer communication are rehearsed. Optional C05 optimization, remote assistance and I09 roadside sensing stay out until separately gated.
G2 constraint register: ODD = approved operating conditions (road, speed, weather, time, etc.). DDT = real-time operational / tactical functions needed to operate in traffic (including OEDR); record which actor performs them. Define DDT fallback separately by actor, trigger, response and intervention boundary. Also record operating limits (hours / staffing / recovery), infrastructure (node power / energy / footprint) and business limits (fare / demand / transit overlap). For each, record variable, unit / value or TBD, source URL, owner, affected KPI and PL line. Set thresholds only after evidence and ownership are confirmed. Source: SAE J3016_202609, doi:10.4271/J3016_202609.

Interaction concept for user app and operator workspace

A simple request–offer–service flow exposes meaningful choices. Detail appears when the user compares or changes a plan.

04 / UX FLOW & PROTOTYPE

RESIDENT / VISITOR APP • CONCEPTUAL

01 REQUEST	02 OFFER	03 SERVICE
Plan a journey	**Choose a feasible plan**	**Track and get help**
Origin / destination + pickup point	Activity / trip + sharing option	Booking / mission state + age
Time window + allowed detour	Pickup / arrival + detour limit	ETA + last update
Party + access + luggage needs	Seat / access + cargo compatibility	Pickup / onboard / arrival proof
Consent + sharing preference	Total price + discount / refund	Help, cancel or recovery path
Validate → submit intent	**Accept plan / keep original**	**Confirm outcome / recover**

CHOICE RULE **Keep the original plan visible. Confirm timing, total price and cancellation terms.**

OPERATOR WORKSPACE • ACTION CONTEXT

CAPACITY & QUEUE

Demand by zone / time
Reachable vehicles + ready time
Seats / cargo space per segment
Hub / loading slots and queues
Committed trips + spare capacity
Cases by severity / dueAt
Passenger recovery reserve
Cargo custody / failed handover
Throttle infeasible new offers

WHY THIS OFFER

Received plan + consent scope
Capacity / evidence age
Original and inserted route
Pickup / arrival / detour impact
Discount + freight revenue
Dwell, route and handling cost
Passenger / cargo rejection reason
Original choice stays accessible
No feasible capacity: no promise

PERMITTED ACTION

Recheck all route segments
Protect accepted passenger limits
Hold seats / cargo / loading slot
Confirm terms and custody owner
Reconcile timeout / lost ACK
Audit each route revision
Release only cleared capacity
Recover each passenger party
Own return / failed delivery

USABILITY TASKS TO TEST

Test optional data access, original choice, revised timing and total cost. Measure success, decision time, comprehension, opt-out and acceptance by accessibility need. Include refusal, stale capacity, cancellation after positioning and lost ACK.
Users must identify the committed plan, decline while retaining the original option and reach help. Operators must explain infeasibility, protect commitments and recover without duplicates. Record critical errors and failed recovery separately.

Prototype evidence: decision time, price / timing comprehension, recovery completion and reasons for refusal. No forced consent or automatic schedule change.

STATE MODEL

DRAFT → QUOTED (expiresAt) → PENDING (hold) → CONFIRMED (commit) → IN_SERVICE → COMPLETED (actuals + passenger outcome) / CANCELLED (reason + release). Timeout / lost acknowledgement → UNKNOWN → query and reconcile; never duplicate a side effect. Revision conflict → refresh terms; stale evidence → block offer or action with reason.

Five mechanisms, service guardrails and net value

Evaluate information effects and transport sharing against matched baselines, with separate passenger, shipment and vehicle measures.

05 / SUCCESS MEASURES

NORTH STAR

Verified completed journeys

count only when mission evidence + service actual + user outcome reconcile

01 POSITIONING

Pickup wait

max(0, actual pickup − requested pickup time). Median / p95.

C02 request + C04 arrival
Report late and missed pickup separately.

Empty travel

Empty vehicle-km / completed trip. Include advance positioning.

C04 movement trace
Include failed positioning and returns.

Dispatch success

Assigned missions / eligible requests. Retain capacity rejects.

C02 request + C04 assignment
Assignment is not completed service.

Required fleet

Minimum tested fleet meeting all service gates, with uncertainty.

Fleet sweep + P25–P28
Hold demand / hubs / service gates fixed.

02 NEW DEMAND

Offer conversion

Accepted activity offers / eligible offers shown.

C02 offer and acceptance
Deduplicate repeat offer impressions.

Additional trips

Treatment − matched baseline completed trips. Identify substitution.

Intent linkage + comparison arm
Net change includes displaced use.

Fleet utilization

Productive / available hours. Also report scheduled hours.

C04 + fleet schedule
State productive-hour scope explicitly.

Activity use

Verified additional provider visits. Keep gross and net use separate.

Provider actuals + comparison
Separate visits from user spending.

03 SMOOTHING

Shift acceptance

Accepted time changes / eligible shift offers shown.

C02 offer + booking revision
Retain refusals and expired offers.

Peak demand

Requests per fixed interval before / after accepted time shifts.

Original and revised request time
Fix interval width across all runs.

Unserved demand

Unserved eligible requests / all eligible requests by zone and time.

C02 + final request disposition
Show cause, zone and peak interval.

Promise kept

On-time pickups / pickups due. Include missed pickups.

Accepted window + C04 arrival
Report original-window change too.

NET ECONOMICS

Net revenue

Recognized revenue after discounts / refunds, counted once.

C07 contract + ledger
Keep operator / provider scope distinct.

Incremental cost

Extra travel, energy, service and app operating cost.

C04 / C05 + cost ledger
Include added support and refunds.

Incremental contribution

Δ recognized revenue − Δ attributable cost. Report total and per completed trip. Include app cost once and document allocation.

Matched arms + C07 ledger
Show total and unit economics.

Interaction effect

Combined gain − sum of separately tested gains. Report overlap and capacity effects.

Paired runs / controlled pilot
Gains share one baseline and unit.

PAIRED ESTIMATE $\Delta K = K_{effect} - K_{baseline}$. Report repeated-run differences and uncertainty.

04 / 05 SHARED-CAPACITY MEASURES

Passenger output, freight delivery and net contribution

Passenger journeys / vehicle-hour; seat-km utilization; detour; freight delivery / lateness; cargo dwell; displaced passenger trips. Detail: P29–P30.

MEASUREMENT CONTRACT

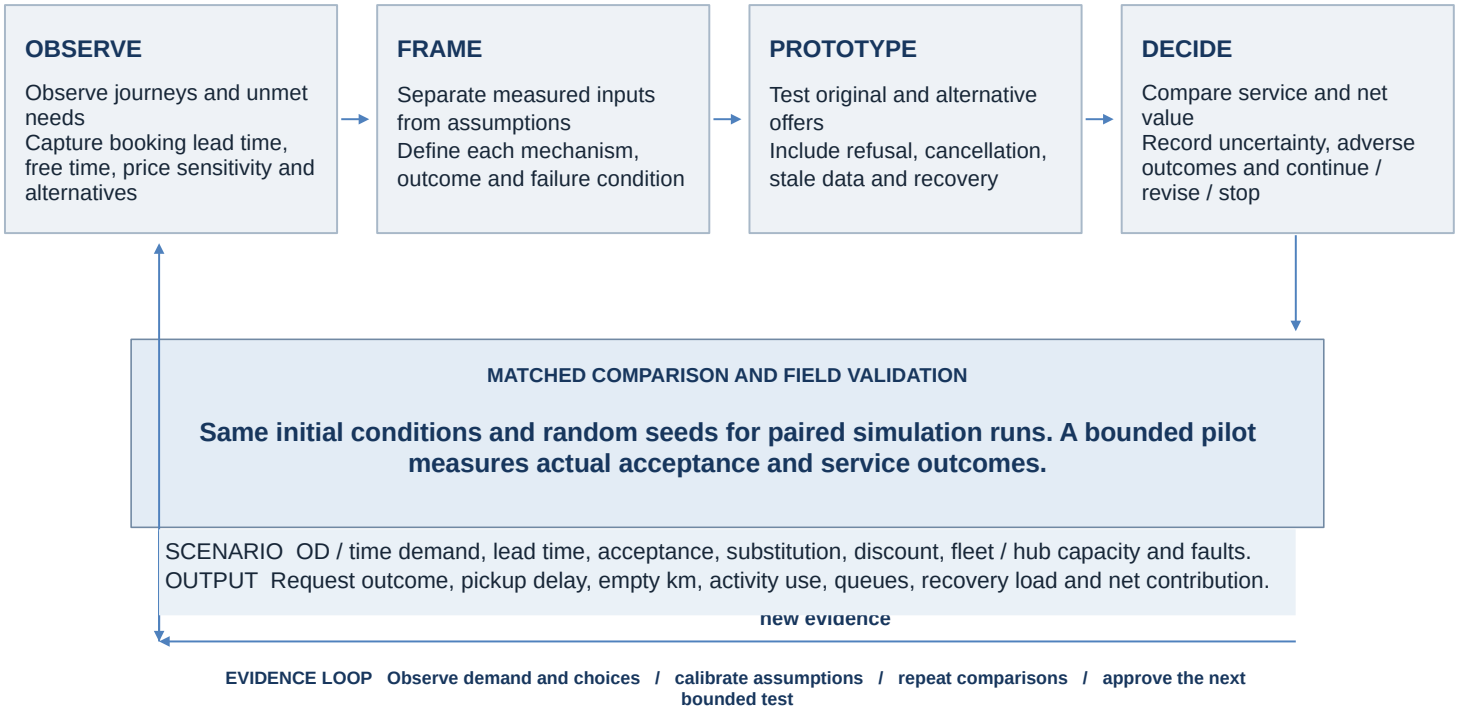
Service gates apply to each passenger and shipment. Separate booking counts, people, deliveries and vehicle missions. Targets require baseline evidence.

ATTRIBUTION: separate additional demand, time shifts, substitution, pooled passengers and delivered shipments. A shared route has several intents but one vehicle mission. Count route distance and common costs once. Effects 01–03 use matched inputs. Compare 04 / 05 in four conditions, then test interactions.
SERVICE: retain unserved passenger and cargo requests. Check pickup / arrival promises, detour, seats, payload, volume, hub queues and recovery. Report passenger and freight fulfillment separately. Confirm observed use and custody outcomes.
FINANCE: net passenger fares already reflect sharing discounts and displaced trips. Add recognized freight revenue, then deduct route, handling, energy, support, cleaning, recovery, returns and app costs once. Show total contribution alongside unit measures. Keep CAPEX and project cash flow distinct.

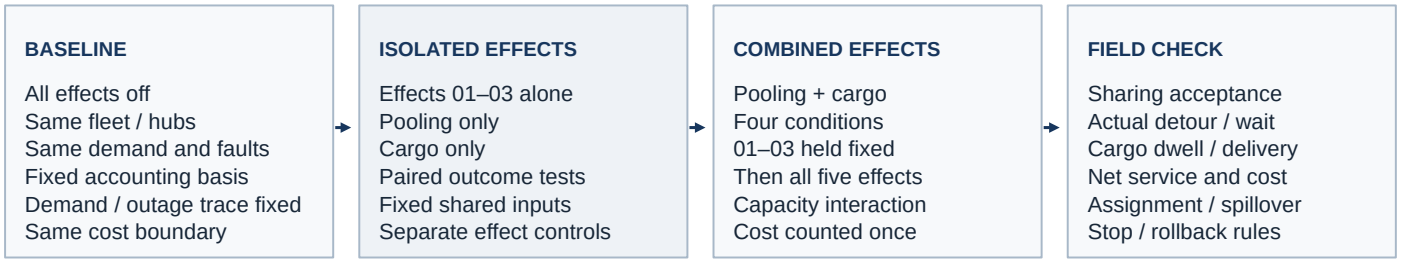
Controlled experiments and evidence workflow

Simulation tests five mechanisms under stated assumptions. Field trials measure actual acceptance, shared-route performance and handling time.

06 / DISCOVERY & VALIDATION



EXPERIMENT DESIGN / ATTRIBUTION AND INTERACTION



STRESS TESTS

Sweep lead time, forecast error, acceptance, substitution and discounts. Vary route overlap, party size, allowed detour, cargo mass / volume, handling time and delivery windows. Test peak demand, hub loss, no-shows, absent recipients and vehicle recovery. Preserve passenger / shipment identity. Reject future-information leakage. Gains fail when realistic costs erase contribution or any accepted service limit breaks.

Proposed tests only. Simulation outputs do not establish field impact or guarantee service-level attainment.

REPRODUCIBLE EVIDENCE RECORD

Record engine / scenario version, input units / sources, demand trace, information receipt times, initial fleet / hub state, seed and effect switches. Retain offer / intent linkage, acceptance, original / revised time, trip / activity actuals, rejected demand and ledger entries. Report paired differences and uncertainty across repeated runs.
Field design: predefine assignment unit, observation window and thresholds. Shared vehicles create spillover across users and zones. Check non-participation, displacement and rebound demand. Keep simulated acceptance distinct from measured behavior.
Decision packet: input provenance, reproducible run, outcome distribution, guardrail failures, sensitivity, accountable owner and a documented continue / revise / stop decision. Unsupported effects stay unvalidated.

Roadmap from discovery to repeatable deployment

Each gate adds evidence for customer response, operating feasibility and net value before the service expands.

07 / RELEASE & SCALE GATES

G1 / DISCOVER

Select users, zones and time windows. Observe lead time, unmet demand, activity alternatives and willingness to shift. Establish traceable service and cost baselines. Record sampling gaps and consent boundaries.

EVIDENCE Journey / demand map, request denominators, alternatives and constraints.
CHECK Missing demand and selection bias.
OWNER PM + Operations.

GO / HOLD / REVISE

G2 / DEFINE

Define five mechanisms and separate controls. Prototype sharing, detour, price and cargo terms. Set passenger and freight gates, cost boundaries and comparison design. Agree consent, capacity and custody ownership before evaluation.

EVIDENCE Causal model, offer tests, KPI dictionary and comparison design.
CHECK Consent, cost boundary and failure criteria.
OWNER PM + Design + Finance.

GO / HOLD / REVISE

G3 / PILOT READY

Pass C01–C07 normal and fault paths. Verify seat / payload limits per segment, route revisions, no duplicate intents and auditable costs. Staff passenger recovery and cargo handling. Rehearse no-show, failed delivery and rollback.

EVIDENCE Acceptance traces, finite capacity, ledger and staffed recovery.
CHECK Fault injection and rollback rehearsal.
OWNER Engineering + Operations.

GO / HOLD / REVISE

G4 / LEARN

Compare individual and combined effects with matched inputs. Measure sharing acceptance, detour and handling time. Retain displaced passenger trips and failed deliveries. Check service gates and net contribution under realistic conditions.

EVIDENCE Effect range, adverse outcomes, full costs and counter-evidence.
CHECK Net gain survives realistic response.
OWNER PM + Operations + Finance.

GO / HOLD / REVISE

G5 / REPEAT

Repeat at a second site with local demand, provider capacity and costs. Confirm gains survive weak acceptance and supply loss. Package configuration, training, monitoring and recovery. Approve expansion within a verified operating envelope.

EVIDENCE Second-site results, support cost, sensitivity and partner acceptance.
CHECK Local limits preserve service and value.
OWNER Rollout lead + site owner.

GO / HOLD / REVISE

HOLD / REPLAN RULES

Schedule alone never passes a gate. HOLD for unresolved safety / privacy / legal risk, unknown execution, missing evidence, unstaffed support, partner unreadiness, unreliable data or unreconciled cost.

REVISE with named owner, required evidence, deadline and temporary scope restriction. Return to an earlier gate when new evidence changes the user problem or service boundary. Record rollback trigger and resume authority.

GATE TRANSITION RULE • Gates are evidence checkpoints, not fixed calendar phases. A failed condition returns to the earliest gate whose assumption is invalidated; scope stays held until the named owner closes the evidence gap and the decision is recorded.

PRODUCT DECISION OWNERSHIP

PM owns user outcomes, scope and KPI decisions. Design owns comprehension and accessibility. Engineering owns contracts, trace quality and release. Operations / partners own service capacity and recovery. Finance verifies revenue scope, cost allocation and investment assumptions. Every HOLD has a named closure owner.

DECISION BASIS Service feasibility and customer choice remain binding even when aggregate contribution improves.

TECHNICAL EVIDENCE IN THE FOLLOWING SYSTEM DESIGN

Reservation and service contracts: slides 8–11 • Remote operation and roadside interfaces: slides 13–23 • Fault injection and release dependencies: slide 24

Capacity, replenishment, recovery and verification: P25–P28. Shared rides and passenger–freight rules: P29–P30. All tests remain proposed.

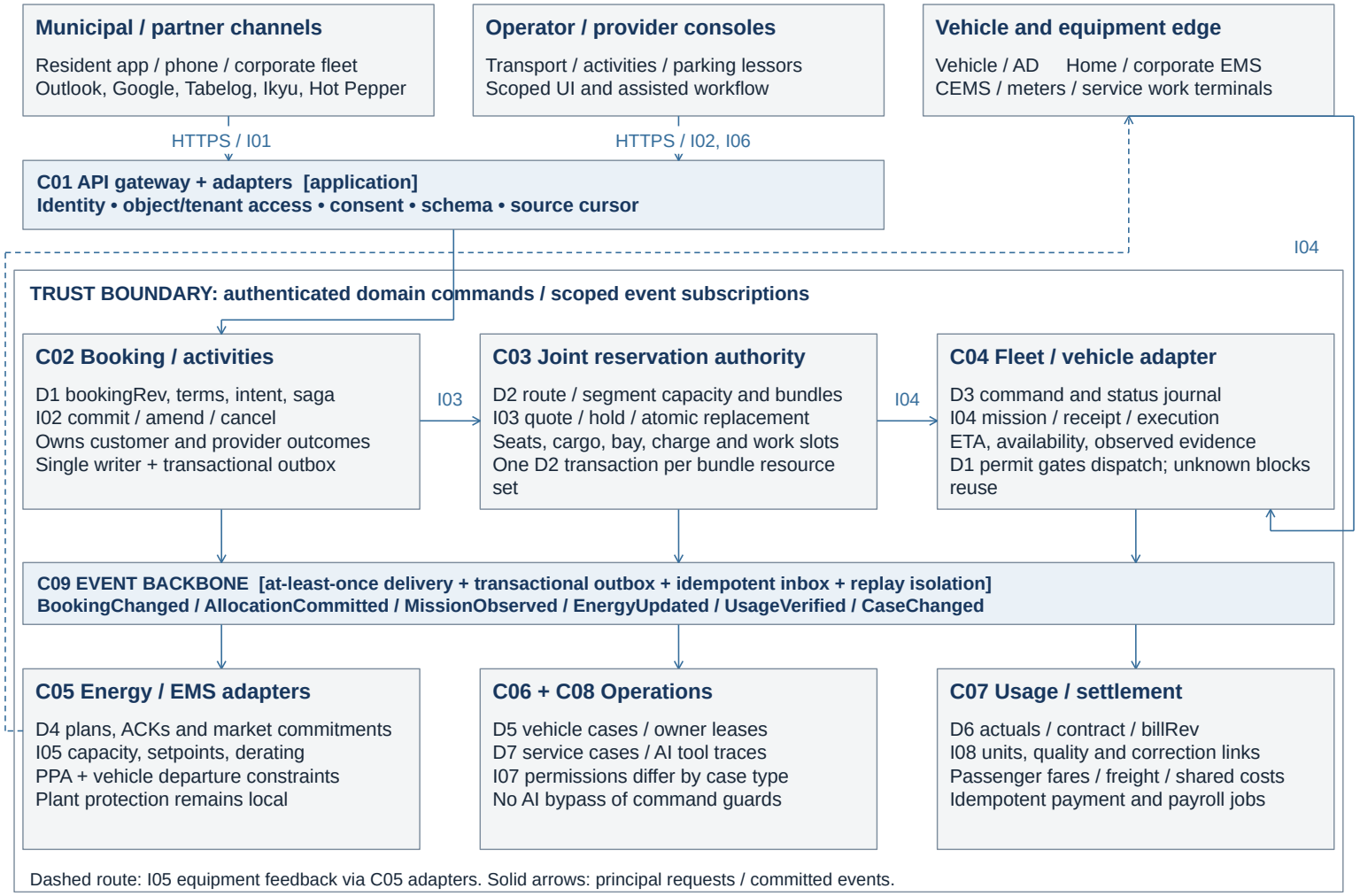
GO / HOLD RECORD

Decision + date • user segment / site • baseline and observed KPI • acceptance trace • unresolved safety / legal / operational risk • scope restriction • accountable owner • next review trigger. GO requires named service owner, trained operation, support coverage, data quality and rollback path. Finance record: CAPEX and OPEX by node and project; node and project payback; IRR; sensitivity range; Variable / Value / Source URL / formula-use link for every assumption.

Container architecture and deployment boundaries

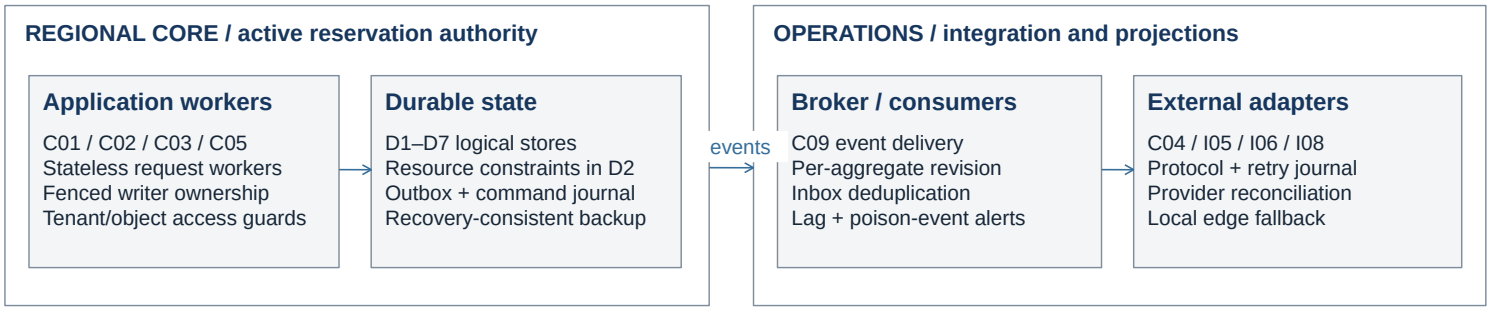
Proposed common PaaS for integrated reservation, dispatch and energy operations. Diagrams define candidate boundaries and contracts. Desk review is complete; implementation and validation remain pending.

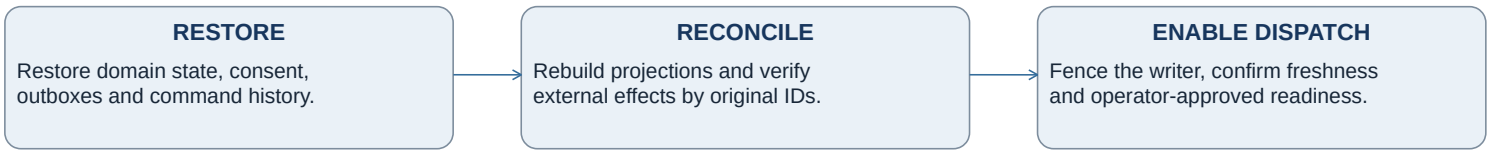
1.1 Logical containers, request paths and event distribution



Dashed route: I05 equipment feedback via C05 adapters. Solid arrows: principal requests / committed events.

1.2 Runtime deployment, isolation and recovery





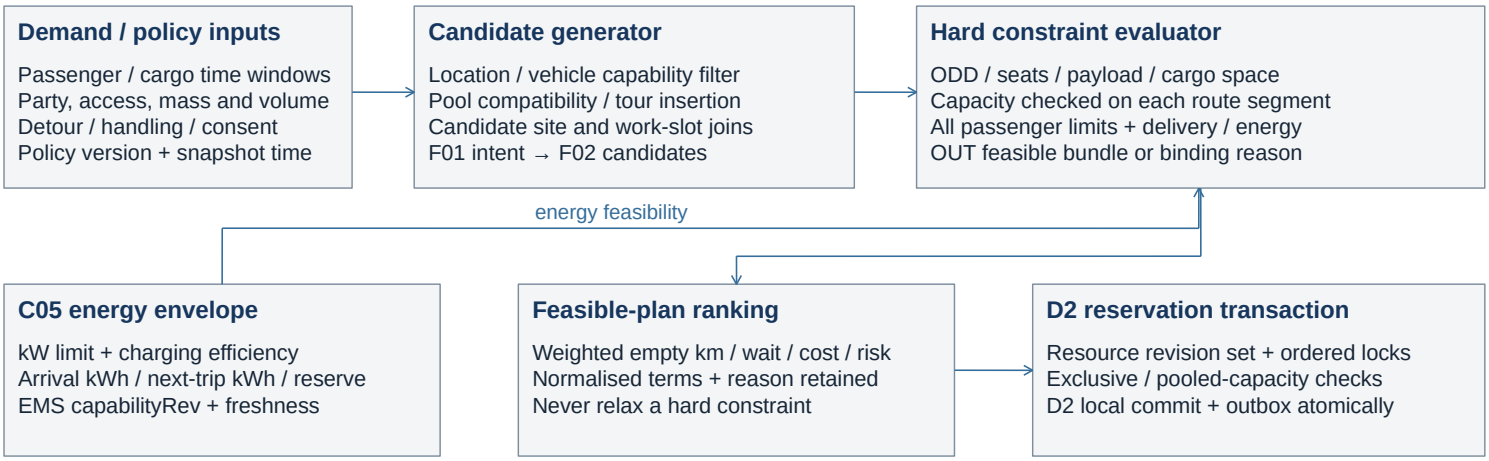
Connector failure may leave local intake available; confirmation still requires all mandatory dependencies. Stale vehicle/site state blocks affected capacity. Replay must use separate projections and idempotent effect handlers.

G2 decisions: hosting zones, failover mechanism, protocols, RPO/RTO, latency and freshness budgets. The primary service integrates booking, dispatch and energy. Equipment/JV/eVTOL options attach through capability interfaces.

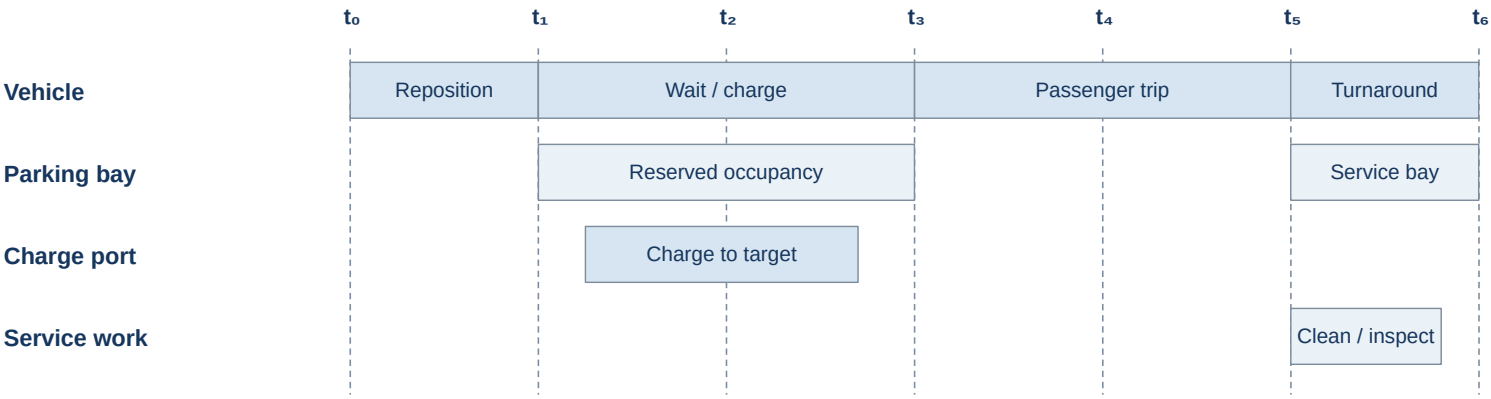
Joint scheduling, resource timing and states

Feasibility covers shared-route insertion, passenger and cargo dwell, turnaround, replenishment and the next departure. All accepted passenger limits remain binding.

2.1 C03 component decomposition and constraint propagation

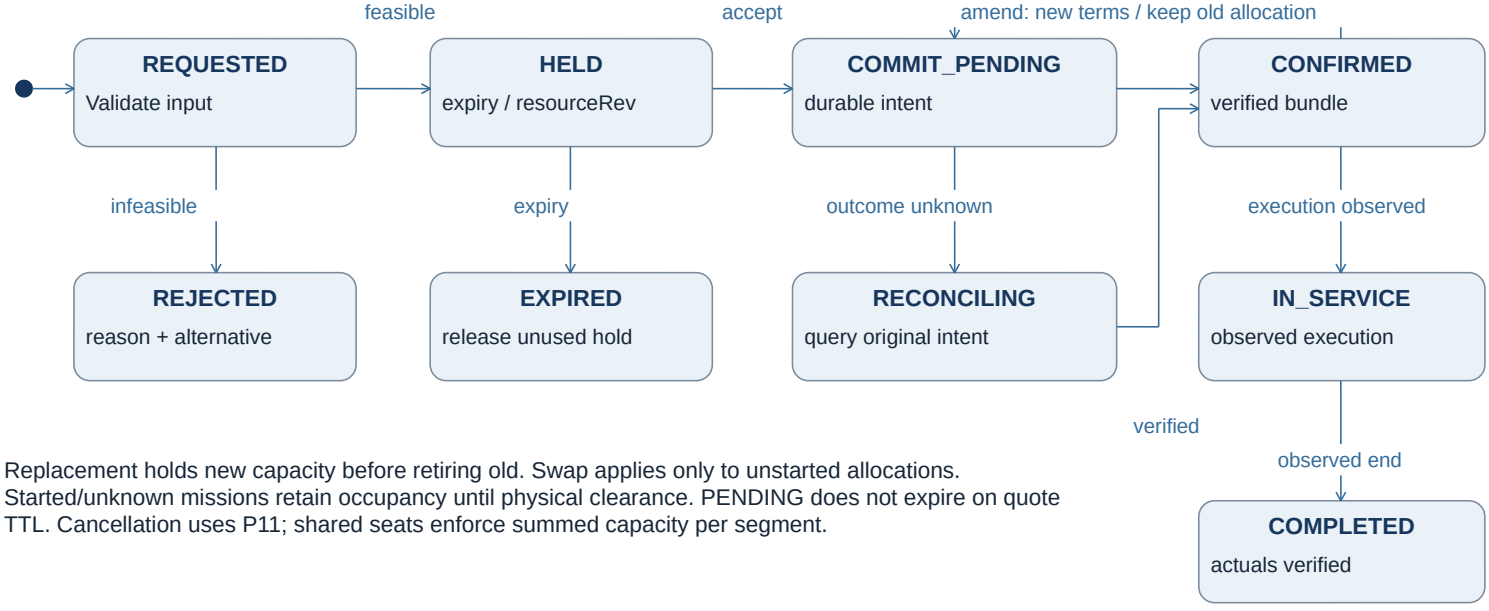


2.2 Coordinated timeline: one vehicle, one bay and one charging session



Departure guard: all required tasks finish before departure; precedence permits only approved overlap. Electrical branch: $E_{dep} = E_{arr} + \Sigma(\eta_{ch} P_{AC} \Delta t_h) - E_{aux}$, with $E_{trip} + E_{reserve} \le E_{dep} \le E_{usable,max}$. kW × hours = kWh.

2.3 Selected reservation transitions and replacement constraints



Replacement holds new capacity before retiring old. Swap applies only to unstarted allocations. Started/unknown missions retain occupancy until physical clearance. PENDING does not expire on quote TTL. Cancellation uses P11; shared seats enforce summed capacity per segment.

Shared rides and cargo require segment capacity, route revision and distinct booking outcomes (P29–P30). Count the vehicle cycle once, including empty travel, boarding, cargo dwell, turnaround and energy. Reserve loading bays, staff and hub capacity separately (P25–P26).

Entity relationships and message contracts

Identifiers, cardinalities, revisions and uniqueness constraints preserve the chain from an accepted journey to resource use and settlement. Example payloads define proposed contracts.

3.1 Domain model with keys and ownership

PlanReference / C01

PK tenantId, planRefId
provider, accountRef, sourceEventId
instanceId, sourceRevision
consentId, purpose, expiresAt
startUTC, endUTC, IANA zone
Unique within tenant:
provider/account/event/instance

0..1 : N

Booking / C02

PK tenantId, bookingId
FK planRefId (optional for app input)
bookingRev, termsRev, state
riderRef, pickupWindow, destination
currentBundleId, intentId
CAS: expectedBookingRev

1 : N

ResourceBundle / C03

PK tenantId, bundleId
FK bookingId, replacesBundleId?
bundleRev, state, expiresAt
policyVersion, inputSnapshotRef
intentId, idempotencyKey, result
Unique: tenant/intent/key

1 : N

1 : N

ActivitySaga / WorkOrder

PK tenantId + sagaId / workOrderId
FK bookingId, bundleId?
providerRef, slotRef, staffRefs[]
conditionsRev, acceptedTermsRev
state, owner, dueAt, evidenceRef
Tasks: cleaning / inspection / handover

ResourceInterval / C03

PK tenantId, intervalId
FK bundleId, resourceId
startUTC, endUTC [half-open]
capacityType, quantity, unit
reservationRev, allocationState
Exclusive: no overlap; pooled: sum ≤ capacity

N : 1

Resource / Capability

PK tenantId, resourceId
kind: vehicle/bay/charger/staff
siteId?, operatorRef, contractRef
capabilityRev, status, observedAt
capacity, constraints, quality
Availability needs current evidence

Bundle references below: 1:N Mission commands • 1:N EnergyPlan revisions • 1:N verified usage and ledger entries

Mission / Command / C04

PK tenantId, commandId
FK bundleId, vehicleId
missionRev, routeRef, validUntil
deliveryState, executionState
statusSeq, observedAt, receivedAt
Same commandId: one immutable payloadHash

EnergyPlan / C05

PK tenantId, planId, planRev
FK bundleId, siteId
capabilityRev, powerIntervals[]
departureMinKWh, expiresAt
acceptedRev, executionEvidence
Plan revision ACK ≠ delivered energy

UsageFact / Ledger / C07

PK tenantId + usageId / ledgerEntryId
FK bundleId, contractRev
meterPointId, interval, qty, unit
quality, sourceRev, payerRef
correctionOf?, billRev, currency
No destructive overwrite of actuals

3.2 Command, result and event examples

I03 CommitBundle / command

```
{
  tenantId: "operator-A",
  intentId: "amend-41",
  bookingId: "bk-18",
  expectedBookingRev: 6,
  newBundleId: "rb-52",
  oldBundleId: "rb-37",
  expectedResourceRevs: [104, 28],
  acceptedTermsRev: 3,
  capabilityRev: "cap-9",
  idempotencyKey: "amend-41"
}
```

Same key + different payload:
reject IDEMPOTENCY_CONFLICT.

I03 Result / guarded outcome

```
{
  intentId: "amend-41",
  state: "COMMITTED",
  bundleId: "rb-52",
  bundleRev: 105,
  intentIdForD1: "amend-41",
  committedAt: "<UTC instant>",
  externalReleaseJobs: ["j-4"]
}
```

Other explicit outcomes:
HELD / REJECTED / PENDING
reasonCode + retryability
Unknown ≠ rejection.

Event envelope / all domains

```
{
  eventId: "evt-902",
  type: "AllocationCommitted",
  schemaVersion: 1,
  tenantId: "operator-A",
  aggregateId: "rb-52",
  aggregateRevision: 105,
  occurredAt: "<UTC instant>",
  receivedAt: "<UTC instant>",
  correlationId: "amend-41"
}
```

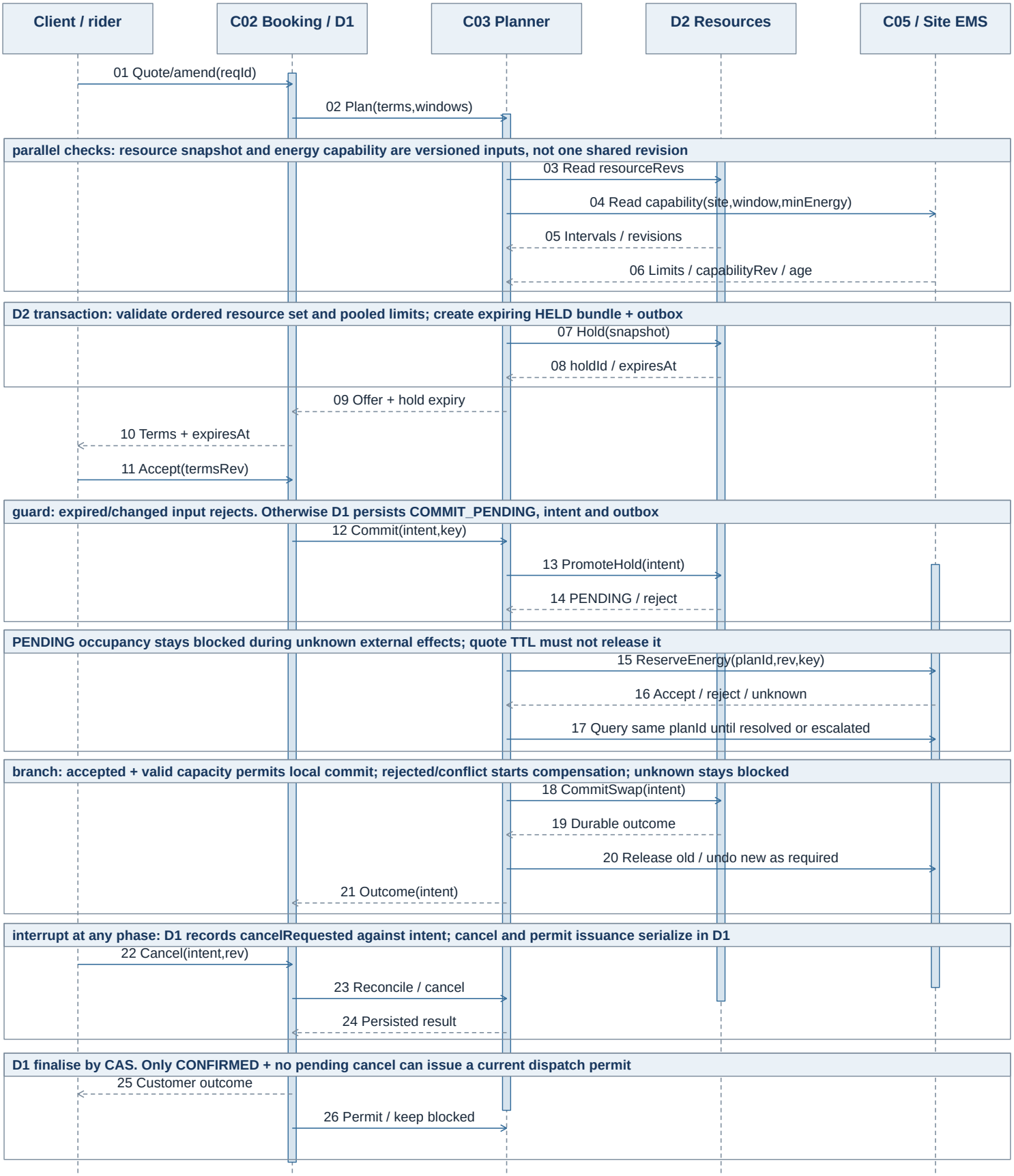
Inbox dedup → revision check
→ apply / fetch gap / quarantine

Keys/FKs include tenant scope; validate references within that scope. expectedResourceRevs maps to the ordered resource set. D1 owns bookingRev. Atomic guarantees cover D2 only. Examples are conceptual payloads, not complete API schemas.

Booking saga, cancellation and dispatch gate

C02 owns booking intent and cancellation. D2 makes only local atomic allocation changes. External EMS effects use a reconciled saga; no distributed atomic commit is assumed.

4.1 Hold promotion, external outcome and authoritative dispatch permit

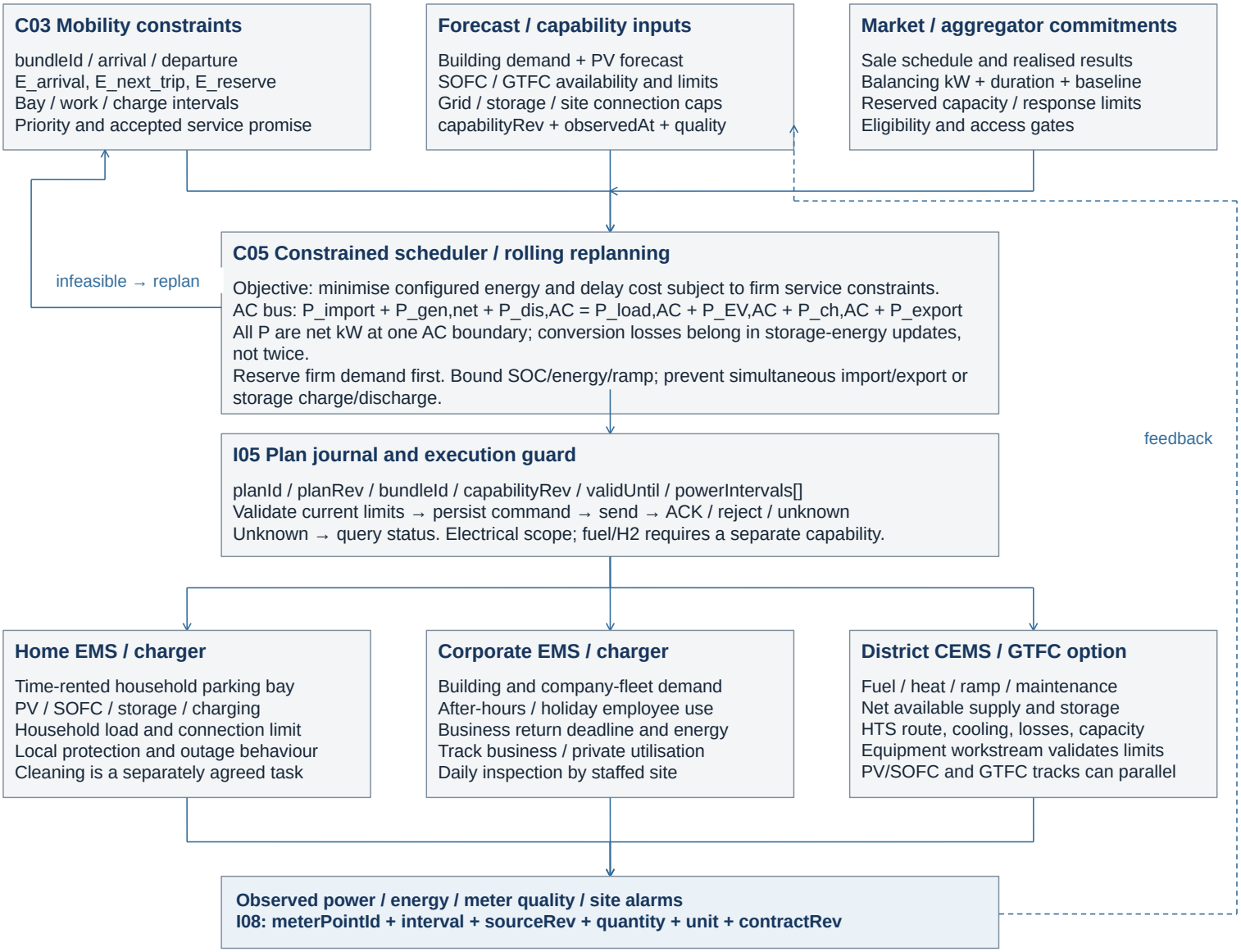


C04 obtains the current D1 permit before releasing a mission. Cancellation after permit issuance becomes cancel/replan of an in-flight mission, not a guarantee of zero movement. Release old physical capacity only after vehicle/EMS clearance. Failed compensation stays owned and reconciled.

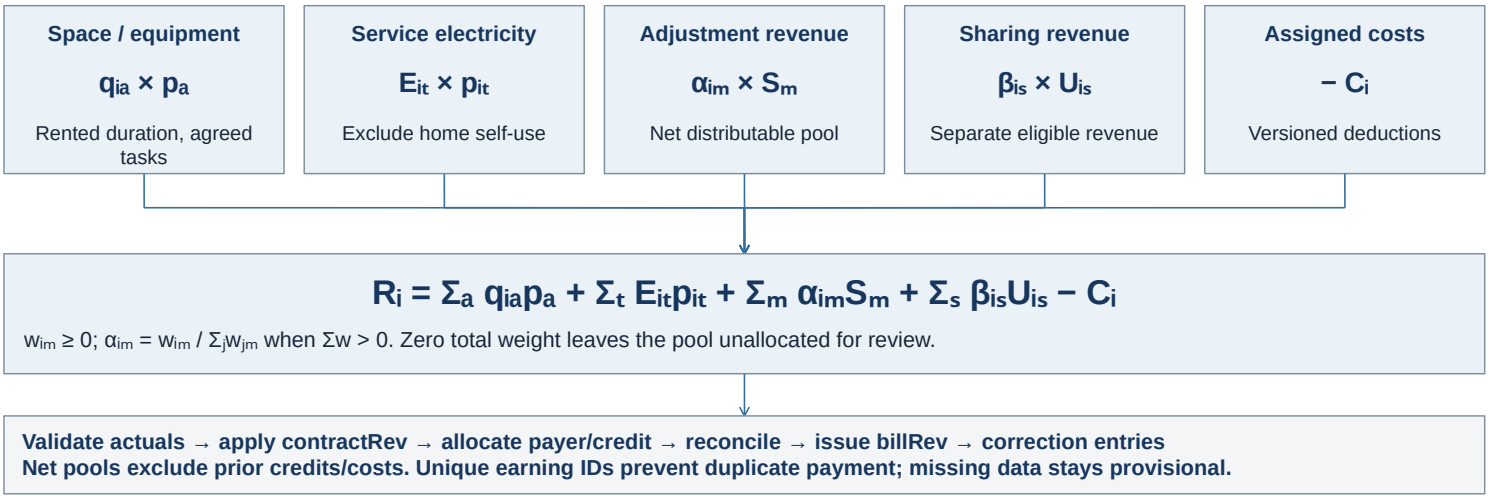
Energy control loops and contribution accounting

The mobility schedule imposes departure-energy constraints. C05 allocates feasible power, reconciles equipment response and sends shortages back to C03. C07 settles measured contributions.

5.1 Hierarchical control and measurement paths



5.2 Settlement dataflow and household contribution formula

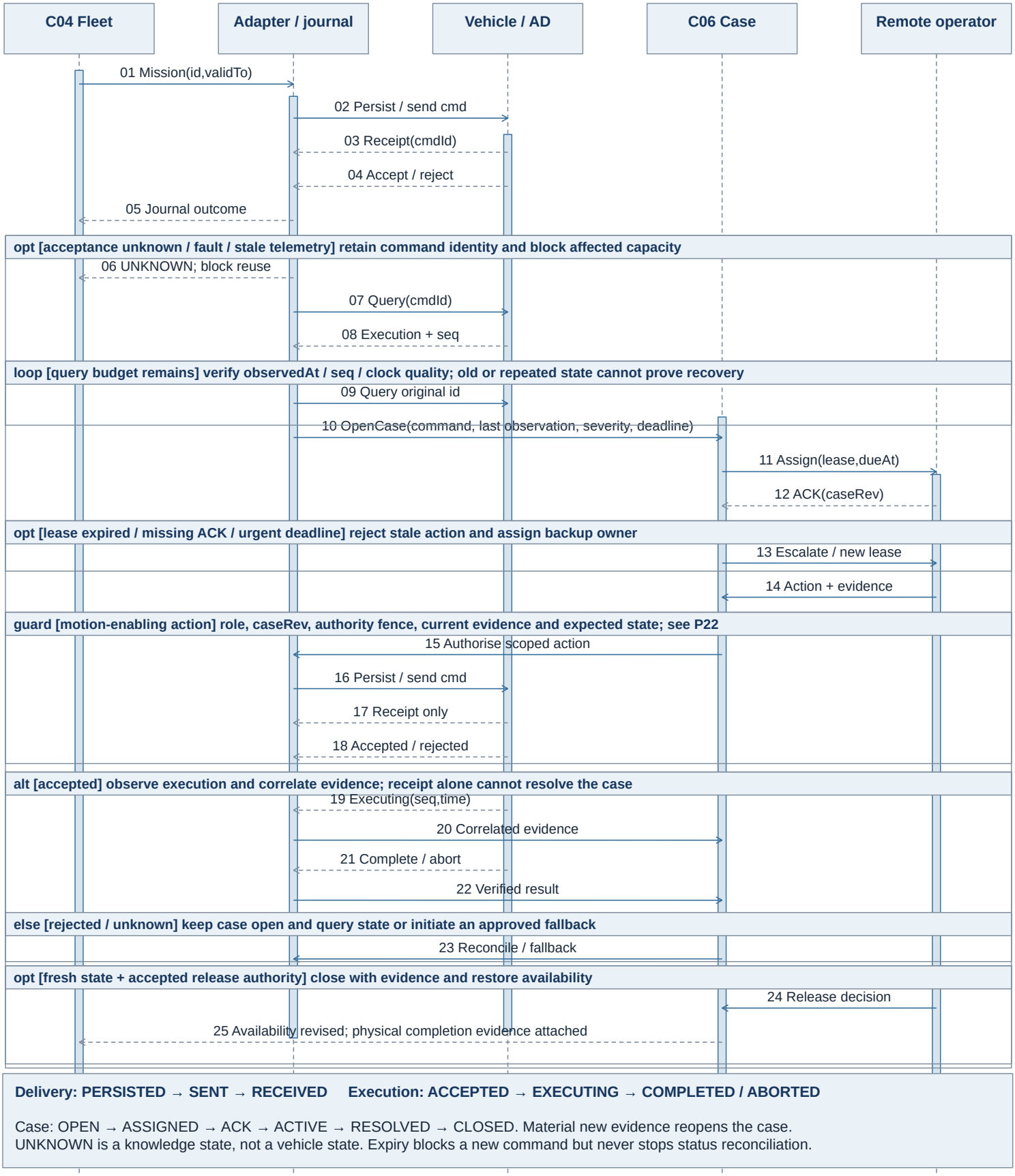


Replenishment queues, site derating and reserve energy constrain each next departure (P26). Settle measured delivery under a versioned contract. Keep net pools, credits and assigned costs non-overlapping. Availability kW and delivered kWh remain distinct.

Remote mission protocol and exception handling

I04 delivery acknowledgment, vehicle execution and I07 case resolution carry different evidence. Permission checks and local vehicle authority constrain every remote action.

6.1 Persistent command, uncertain execution, case ownership and recovery



Delivery: PERSISTED → SENT → RECEIVED **Execution: ACCEPTED → EXECUTING → COMPLETED / ABORTED**

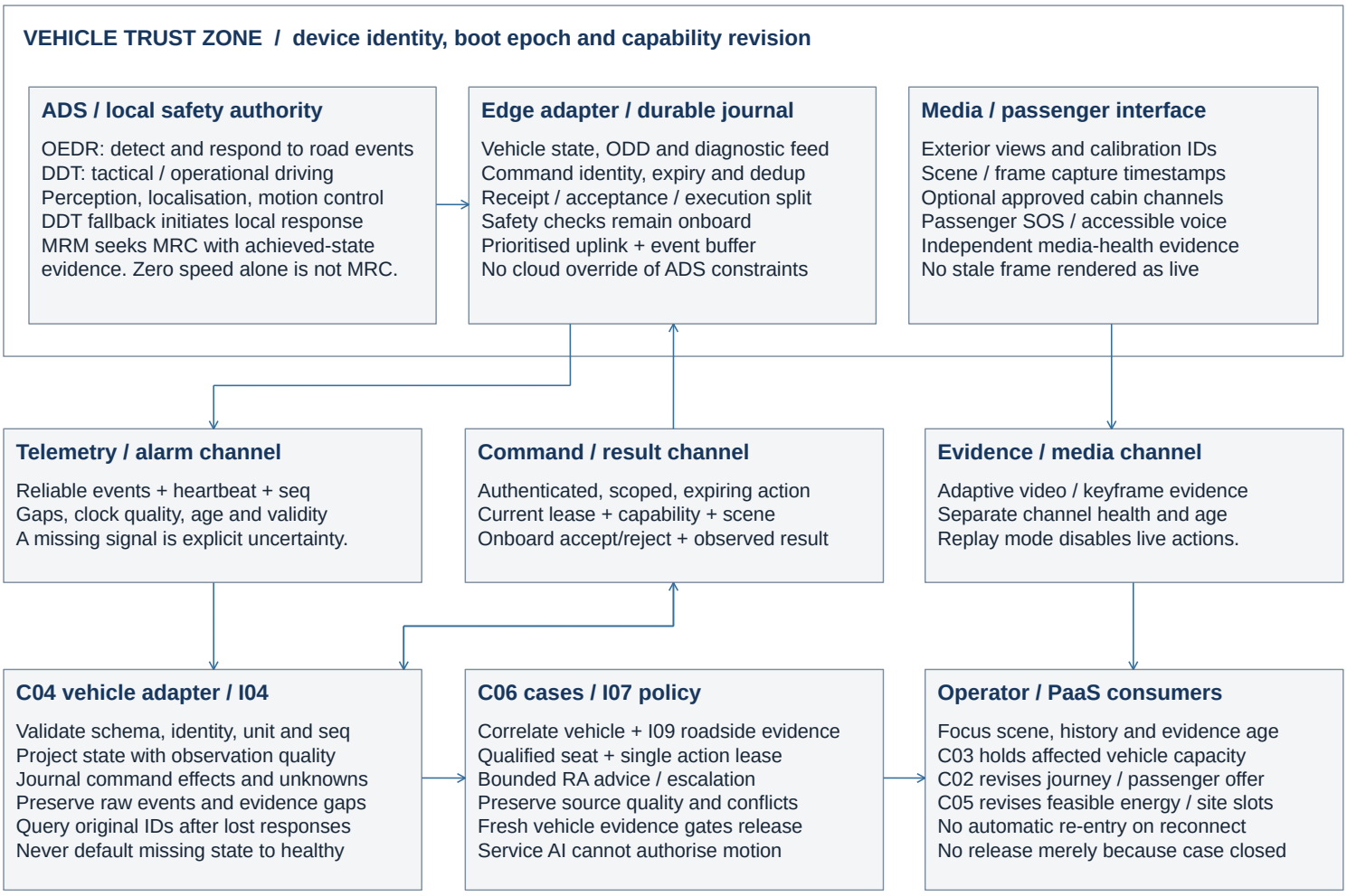
Case: OPEN → ASSIGNED → ACK → ACTIVE → RESOLVED → CLOSED. Material new evidence reopens the case.
UNKNOWN is a knowledge state, not a vehicle state. Expiry blocks a new command but never stops status reconciliation.

Vehicle/ADS owns driving and fallback. Read-only queries and emergency escalation remain available with stale data. Case closure follows P18; motion requests use P22 action-specific guards. Rejected/unknown effects stay reconciled.

Remote architecture and control authority

RM-01 / RM-02. Level 4 ADS retains DDT and fallback authority. Remote monitoring, contextual assistance and passenger support use separate permissions and evidence paths.

7.1 Onboard functions, remote channels and command boundary



Roadside sensors / RSU / edge → C10 I09 validation → C06 monitoring evidence [details P15–P16]

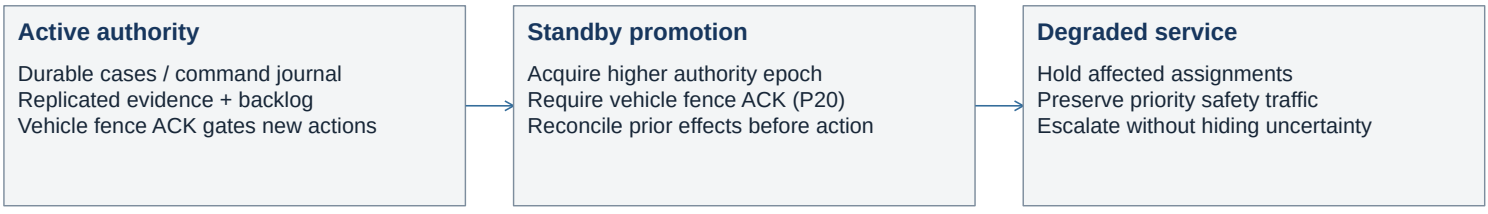
7.2 Operational roles and failure containment

Remote assistance and remote driving

RA supplies contextual advice that the ADS can reject after re-evaluating the scene and ODD. Real-time remote DDT is outside this baseline. Adding remote driving requires a distinct operating envelope, control arbitration, communications safety analysis and operator / approval evidence.

Operational and legal responsibility

Map system roles to the permitted operation, designated supervisor and field-response organisation. Passenger support cannot override door interlocks or authorise motion. Vehicle-specific link-loss and fallback behaviour remains local and does not depend on a remote reply.

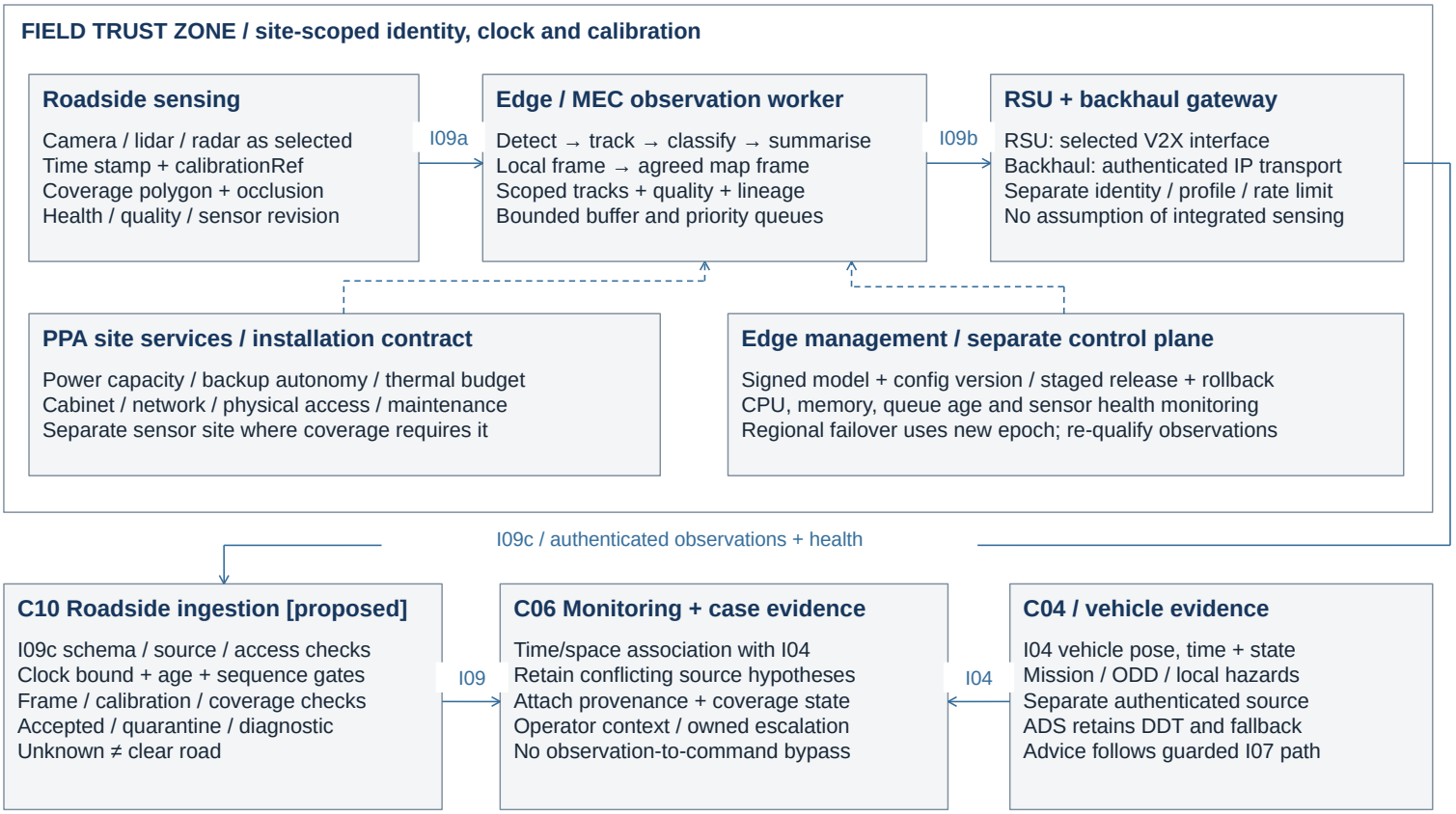


Acceptance: independently partition telemetry, media, primary centre and workstation. Old-epoch requests reject after the vehicle installs its fence. Test durable deduplication and unknown outcomes. Reconnect and case closure require the release predicate on P18 before availability can change.

Roadside sensing and edge computing architecture

RM-13. Roadside observations supplement vehicle evidence. Sensor, compute and communication functions have separate ownership. Proposed I09 feeds monitoring and operational review, not direct actuation.

8.1 Deployment topology and data / management plane separation



8.2 Proposed observation contract and association keys

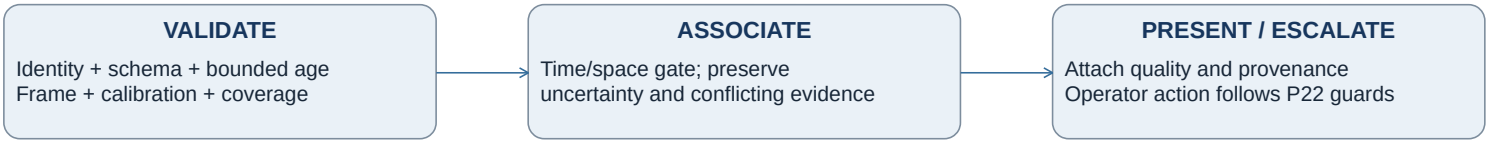
ObservationEnvelope / I09c

tenantId, siteId, sourceId, sensorIds[], bootEpoch, seq
observedAt, emittedAt, clockErrorBound_s, expiresAt
trackKey = sourceId / bootEpoch / localTrackId
frameId, mapRev, calibrationRev, coverageRef
position_m, velocity_mps, uncertainty / covariance
class + confidence, visibility, modelRev, evidenceRef
Signature / authenticated transport + payloadHash

CoverageHealth + association evidence

coverageId / polygon / exclusions / sampledAt
Sensor availability, clock validity, calibration status
Occlusion / weather / queueAge / processing status
Vehicle association: spatial-temporal gate + uncertainty
Candidate link has confidence, method and expiry
No global identity assumed from local track IDs
Multiple reports retain source lineage to avoid double count

8.3 Evidence eligibility, overload and storage boundaries



Invalid / stale / overloaded

Quarantine observations, publish coverage degradation.
Reserve queue capacity for health / critical events.
Drop policy is explicit; historical replay cannot act live.

Evidence / privacy / regional placement

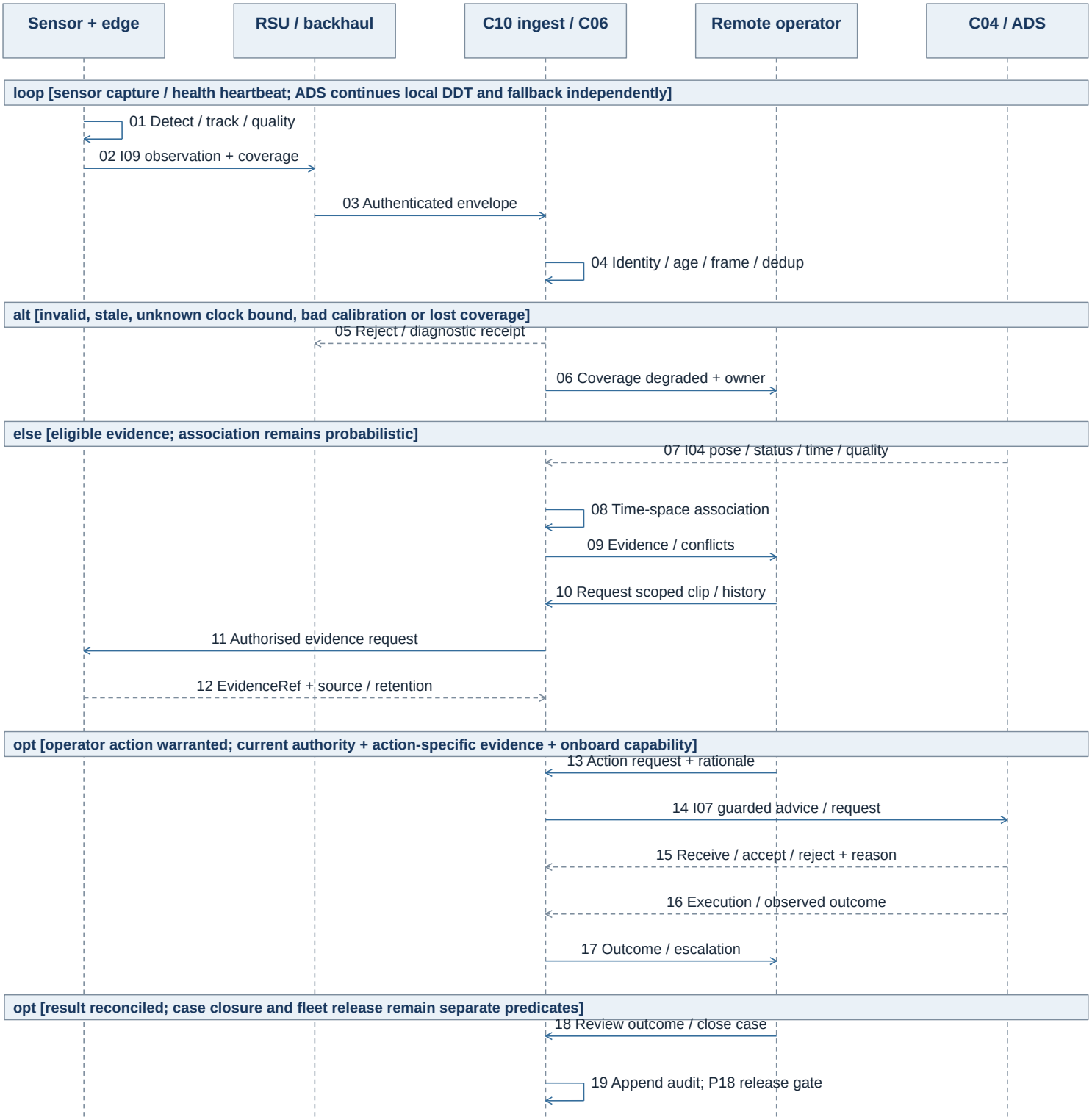
Raw clips by purpose + access + retention contract.
Edge summaries first; authorised evidence on request.
Regional workers scale independently of ADS control.

Thresholds for age, association, coverage and queue size are agreed per use case. CPM mapping and radio profile require a separately verified adapter. Proposed monitoring path does not claim collision avoidance or perception completeness.

Roadside observation and remote monitoring sequence

RM-14. A roadside event becomes scoped monitoring evidence only after validation. The sequence retains independent ADS behaviour, conflicting evidence and unknown outcomes.

9.1 Observation, evidence correlation and guarded response



9.2 Fault branches and planned verification

LINK / EDGE LOSS
Heartbeat expiry → coverage UNKNOWN; retain source time.
No stale frame displayed as current. Reconnect uses new epoch, resync + health checks; replay is historical only.
Test disconnect, reboot, reordering, duplicate delivery.

CONFLICT / OVERLOAD
Keep both source claims; no silent override of ADS state.
Overload reports queue age and dropped observations.
Test occlusion, weather, false association and saturation.
Measure quality, end-to-end age and operator workload.

Telemetry coverage and alarm decision logic

RM-03 / RM-04. Capability-specific signals feed versioned detection and triage. Alarm priorities are operational response classes, not ASIL classifications. Every action uses current, valid evidence.

10.1 Signal families, decision context and missing-data behaviour

ADS / ODD	adsMode, engagementState, oddVerdict/reason, oddProfileRev Route / geofence / weather / illumination constraints as supported.	ODD or mode inconsistency opens a case. Cloud context never widens ODD.
Motion / localisation	pose + coordinateFrame, speed, heading, localisationHealth Covariance and integrity bounds remain distinct supplier quantities.	Detect unexpected stop / deviation. Uncertain pose cannot prove location.
Perception / planning	sensorHealth, occlusionFlags, plannerStatus, blockageReason assistanceRequest + sceneRev + mapRev + frame capture time	Review ambiguity in context. Generic “AI confidence” cannot permit motion.
Chassis / fallback	brake / steering / propulsion health, emergencyStopStatus fallbackReason, mrmState, mrcAchieved, standstill / secureState	Separate manoeuvre progress from achieved MRC. Failed fallback escalates.
Passenger / mission	missionId/rev, pickup/drop-off state, door/ramp/restraint status SOS, occupancy quality, accessibility and handover exceptions	Local interlocks prevail. Unknown occupancy requires supported confirmation.
Energy / link	SOC, usableEnergy, HV/thermal fault, departure reserve margin heartbeat, RTT, loss, jitter, mediaAge, seqGap, clockQuality	Energy shortage revises feasibility. Link loss marks remote state UNKNOWN.

10.2 Evidence validation, alarm lifecycle and response classes

Validate / detect

Identity, schema, unit, epoch/seq
Age / source quality / clock bound
Predicate + persistence + hysteresis

→

Correlate / assign

Vehicle / cause / incident window
Severity + exposure + dueAt
Qualified seat + ACK + lease

→

Act / verify / clear

Guard action and observe outcome
Stable clear condition + release
Reopen on contradictory evidence

P0	Collision / fire / SOS / failed fallback. Immediate escalation, no debounce.
P1	ODD exit / MRM / critical fault / unknown active state. Hold capacity, urgent response.
P2	Assistance / obstruction / boarding failure. Assess exposure and escalate if worsening.
P3	Noncritical service or diagnostic issue. Schedule work under explicit restrictions.

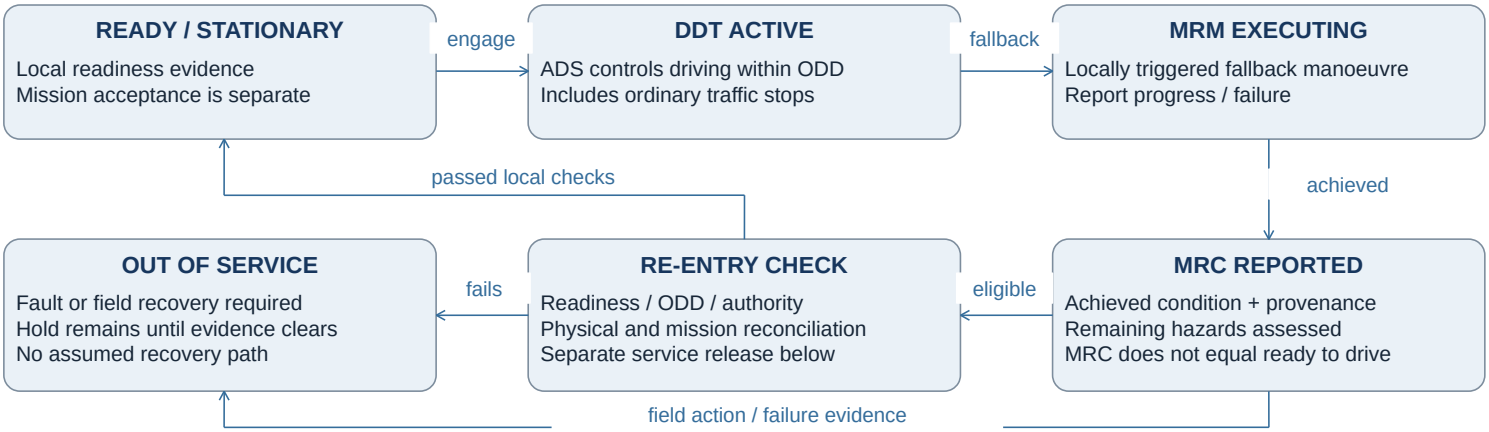
Vehicle I04 / roadside I09 remain separate source envelopes. I09 adds coverageRef, frame/map/calibrationRev, clock bound and source lineage. Preserve original observation time; repeated or replayed samples never refresh it.

Clear only after valid samples span an agreed stable interval. ACK does not clear health. Common-cause grouping retains vehicle exposure and deadlines. Snooze needs reason/expiry and cannot suppress P0. Missing/invalid data raise their own event.

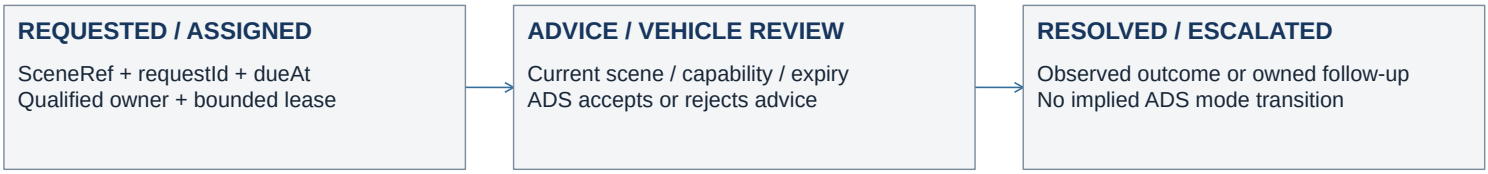
Orthogonal states and guarded return to service

RM-05. ADS operation, assistance workflow, remote knowledge and fleet eligibility are separate dimensions. An ADS can continue its approved local behaviour while a remote assistance request is pending.

11.1 Vehicle-reported operating state: selected transitions



11.2 Assistance lifecycle runs independently of ADS operation



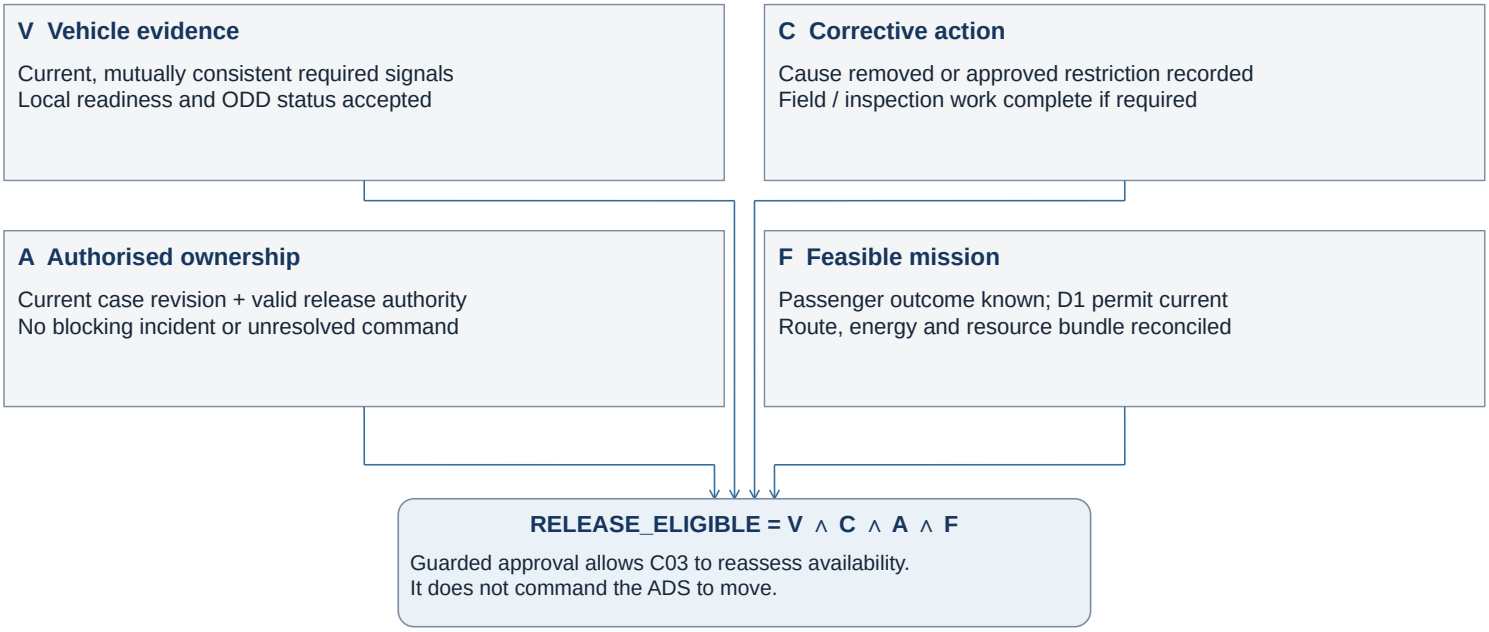
Remote knowledge

CURRENT / DEGRADED / UNKNOWN is per required signal. Fresh data may still report a fault. Unknown physical execution blocks conflicting reuse.

Fleet eligibility / case state

AVAILABLE / RESTRICTED / HELD differs from OPEN / ACTIVE / CLOSED. Closing a case never changes ADS mode or automatically clears an operational hold.

11.3 All release predicates must hold; none substitutes for another

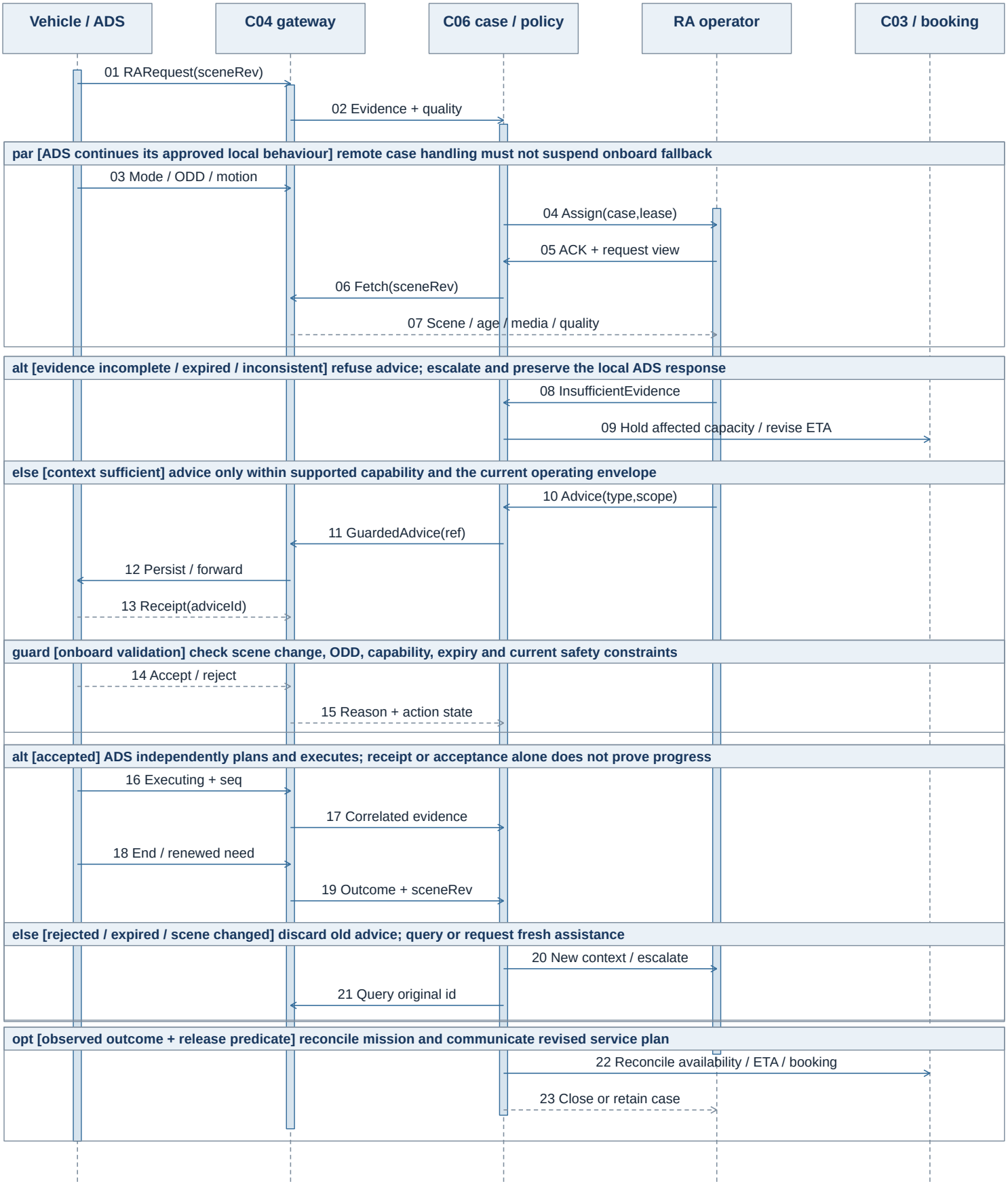


Test cross-products, not only one state chain: DDT_ACTIVE + RA_PENDING; MRC + stale evidence; CLOSED case + critical fault; fresh telemetry + unresolved command. A reported zero speed never suffices for MRC or restart.

Remote assistance: contextual advice sequence

RM-06. Example: the ADS encounters an ambiguous obstruction. A bounded advice transaction carries a scene reference and expiry. The ADS retains trajectory generation, collision checks and motion control.

RM-06A Request, evidence, advice validation and observed outcome

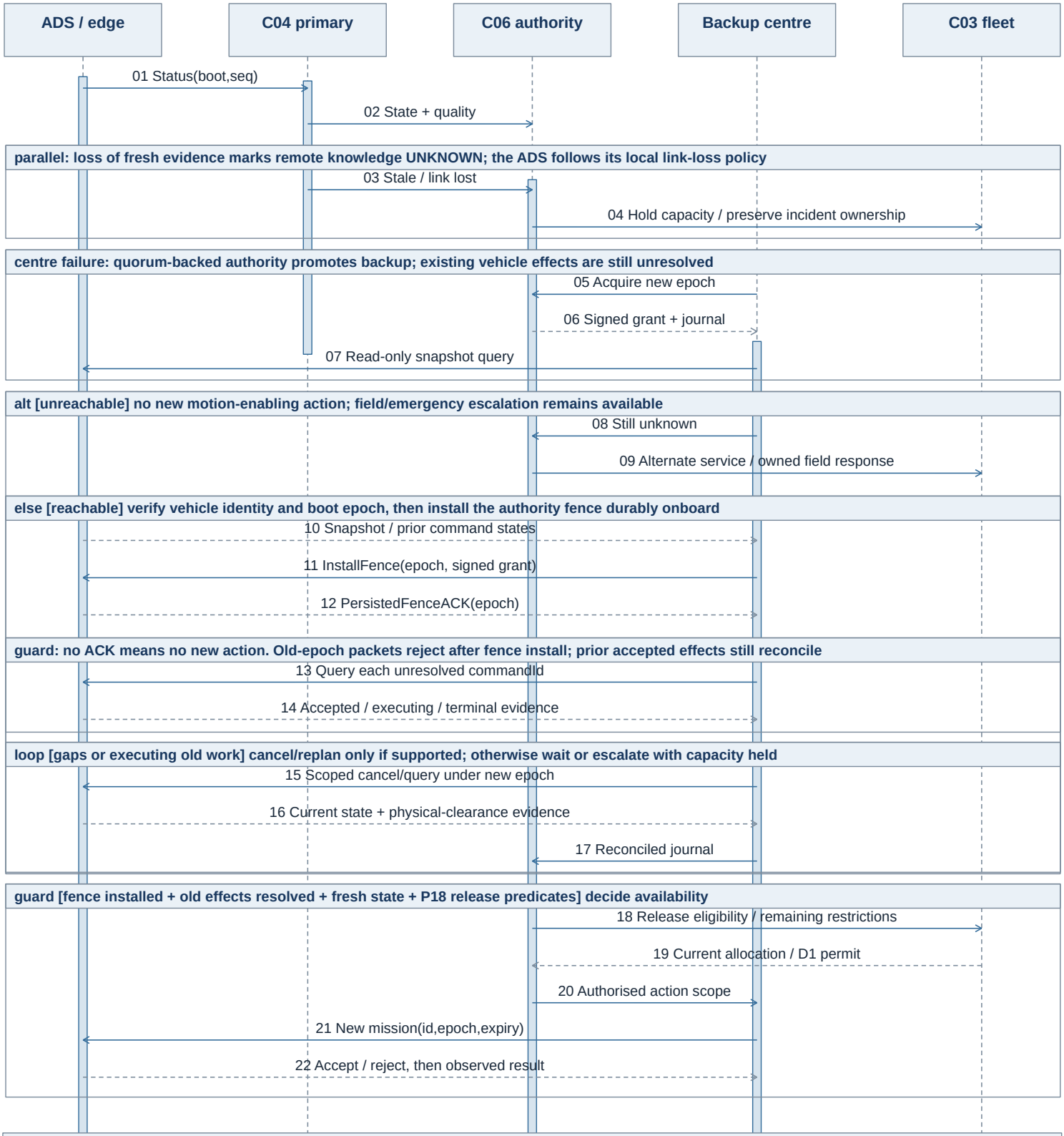


RM-06 invariant: an advice response cannot widen ODD, disable collision avoidance or create a direct actuator path. Timeout/rejection requires reassessment; fresh autonomous resolution can close the case. Executing effects retain their identity.

Link loss, authority fencing and reconciliation

RM-07. Centre authority and vehicle execution require separate evidence. New cloud ownership does not instantly revoke an unreachable vehicle's old authority. Unreachable vehicles remain held.

13.1 Centre change with per-vehicle fence installation



Fence contract: vehicle durably stores its accepted authority epoch and validates signed grants, expiry and command identity. Reboot without recoverable authority/journal evidence blocks new action until resynchronised.
Centre ownership loss blocks new dispatch, but cannot erase an already accepted physical effect. No automatic resend with a new ID. An unsupported fence/dedup capability excludes automated centre handover for that vehicle.

Operator workspace and incident response

RM-08 / RM-09. A focused workspace links fleet triage, ADS evidence, passenger contact and field recovery. Each workstream retains a qualified owner and its own authority.

14.1 HMI wireframe: evidence, authority and operator handover

CENTRE ACTIVE / epoch 42 **Operator O-17 / qualified RA** **Case C-204 / lease current**

Fleet / incident queue

Severity / vehicle / dueAt
Passenger exposure / owner
Oldest unacknowledged case
Common-cause incident link
Seat qualification and health
Escalation always accessible
Colour is not the only cue.

Scene / media / synchronised history

Exterior views / pose / route / scene objects
Map and calibration revision / ADS intended path
LIVE, DELAYED or REPLAY with capture age
Missing view explicit / source quality visible
Align observedAt and receivedAt / mark gaps
Replay disables action. Annotation keeps author, time, evidence reference and case revision.

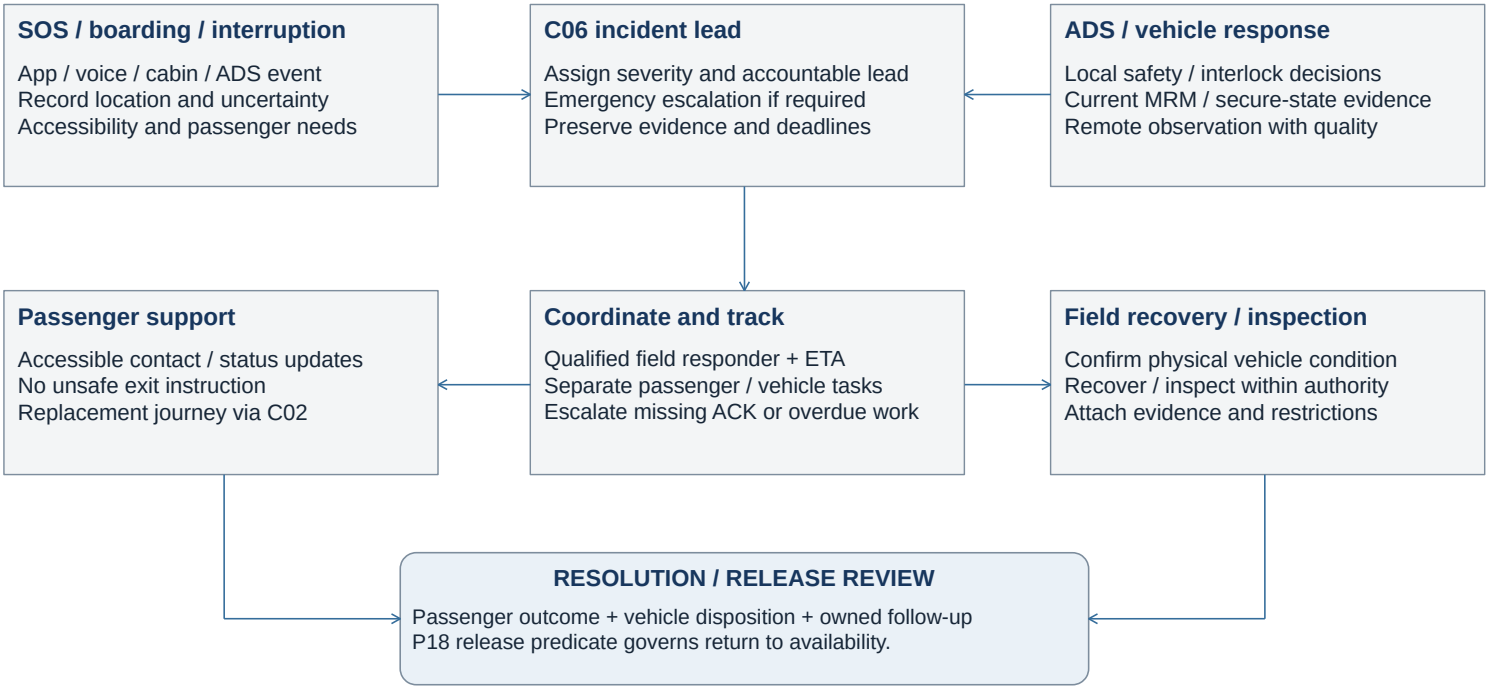
State / permitted actions

ADS mode / ODD / MRM / MRC
Motion, chassis, door, energy
Per-signal age and validity
Assess / advise / stop request
Passenger / field contact
Disabled action shows reason
Scope confirmation before submit

Action and handover record

Vehicle + mission + case revision + lease epoch + reason + advice type + expiry + expected outcome
Action-specific guards use evidence and vehicle authority. Status queries remain available with stale data.
Handover requires acceptance of pending commands, passenger status, field ETA and next deadline.
Expired / unacknowledged ownership triggers reassignment, not silent abandonment.
Relief capacity and supervisor escalation handle concurrent incidents and a failed seat.

14.2 Incident workflow with parallel vehicle and passenger outcomes



14.3 Workload and scenario acceptance

Staffing input: $A = \Sigma(\lambda_k \times E[S_k])$ in matched time units. A/m is a mean-load screen, not a service-level guarantee. Test bursts, breaks, common-cause alarms, handover and centre loss. Include field-team travel and replacement-trip capacity separately. Record detection, ACK, resolution, overdue cases and failed recovery. Restrict new commitments when response capacity is exhausted. P27 defines service-mode and return-to-service gates. Every open branch retains an owner and deadline.

Action-specific guards and time validity

RM-10 / RM-12. Permissions depend on action class. Read-only recovery and emergency escalation remain possible when telemetry is stale. Motion-enabling advice requires current contextual evidence.

15.1 Action class determines its evidence and authority requirements

Read / escalate

Authenticated status or history query
Allowed while state is UNKNOWN
No movement or allocation side effect

→

Request / advise

Current owner, fence and capability
Action-specific scene/state evidence
Stop request uses vehicle contract

→

Onboard decision

Accept/reject with explicit reason
Local ODD and safety checks prevail
No remote steering/throttle baseline

15.2 Proposed command and immutable outcome envelopes

RemoteActionRequest

tenantId, vehicleId, missionId?, commandId
caseId?, expectedCaseRev?, actionType, parameters
capabilityRev, sceneRef?, expectedVehicleStateRev?
operatorId, qualificationRef, authorityEpoch
issuedAt, expiresAt, nonce, payloadHash
reasonCode, evidenceRefs[], approvalRef?

Required fields follow a versioned action schema.
CONTEXT_ADVICE / STOP_REQUEST /
MISSION_CHANGE_REQUEST / STATUS_QUERY
Same ID with changed payload is rejected.

CommandAuditEvent

tenantId, eventId, commandId, correlationId
producerId, bootEpoch, streamId, sequence
observedAt, receivedAt, clock-bound metadata
RECEIVED / ACCEPTED / REJECTED / EXECUTING /
COMPLETED / ABORTED / EXPIRED + reason
stateRevision, evidenceRefs, policyVersion, actor

UNKNOWN belongs to the observer, not the ADS.
Keep raw event + validation result append-only.
Missing terminal evidence stays unresolved.
Expiry blocks new acceptance, not reconciliation.

15.3 Freshness uses bounded clock error and future-time rejection

$\Delta_i = t_{now} - t_{observed,i}$ $U_i = u_{source,i} + u_{observer}$
$ageUpper_i = \max(0, \Delta_i + U_i)$
$timeOK_i = boundsKnown \wedge clocksValid \wedge (\Delta_i \geq -U_i) \wedge (ageUpper_i \leq F_i)$
$valid_i = identityOK \wedge schemaOK \wedge sourceOK \wedge epoch/sequenceOK \wedge timeOK_i$
All terms use seconds. U includes synchronisation error and drift since the last trusted update.
No finite clock bound means time validity is UNKNOWN. Local timers use a monotonic clock.

Response and execution deadlines

$D_{remote} = D_{detect} + D_{queue} + D_{ack} + D_{assess} + D_{transmit} + D_{accept}$. Budget is operational. Do not use it to justify onboard safety deadlines. Expiry uses the receiver's validated time bound; clock uncertainty rejects new time-critical acceptance.

Ownership and release parameters

ADS owner: local link-loss/fallback, action capability, fence and journal persistence. Platform: F_i, clock bounds, sampling, loss, retry and retention. Operations: response class, seat/field coverage and escalation. Each value needs a unit, rationale and boundary test.

15.4 Audit scope and evidence lifecycle

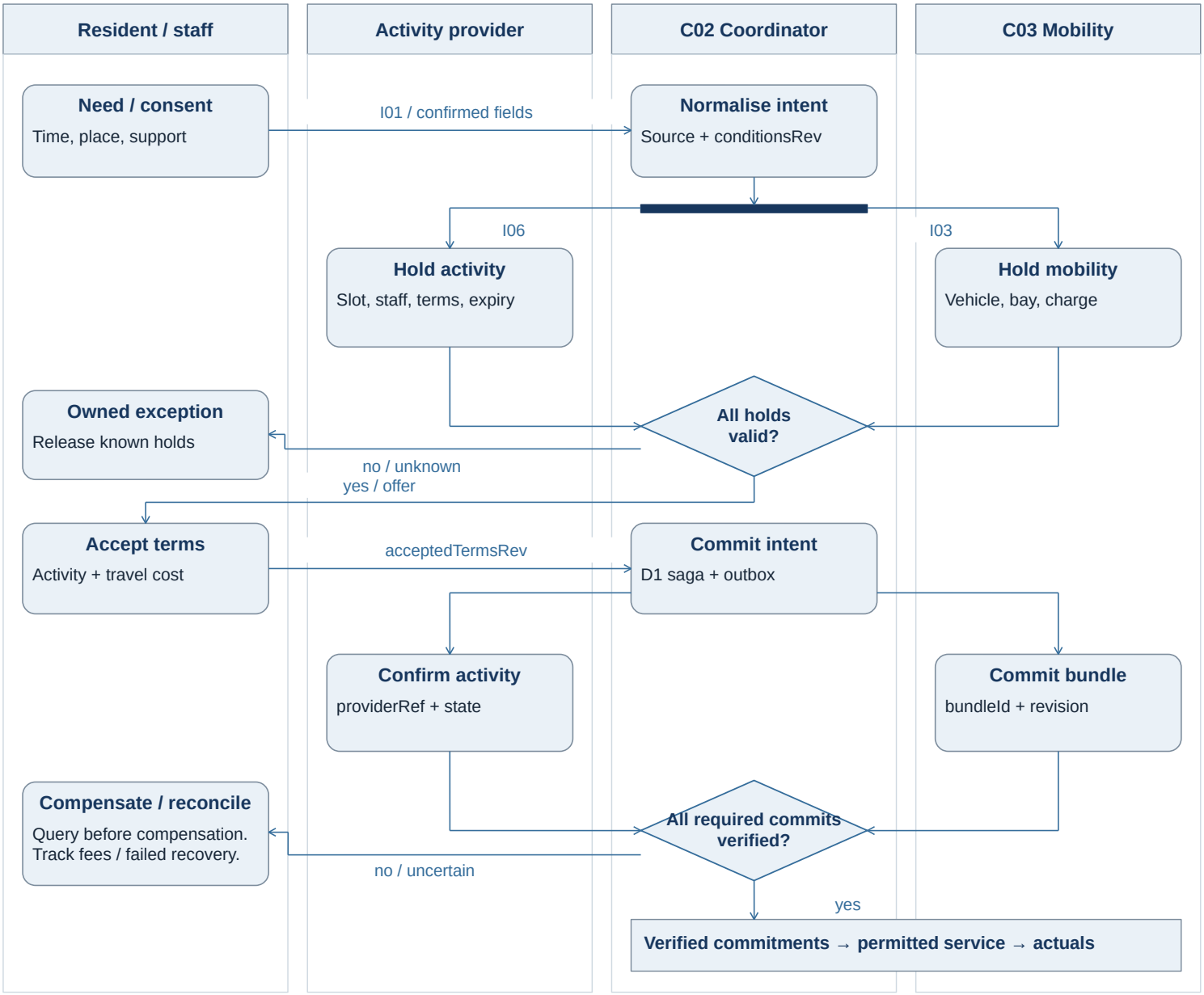
Purpose-limited capture, scoped access, encryption, retained incident evidence and read-only replay. AI may draft or summarise within task scope; it cannot independently authorise ADS movement. Test cross-tenant access, expired authority, future timestamps, unknown clock bounds and duplicate effects.

Desk-reviewed architecture proposal; implementation, performance and safety validation remain pending. Public references support terminology and patterns, not this system's certification. Sources in notes: SAE J3016 baseline, ISO 34503, NHTSA, Waymo, NPA, AWS, Microsoft, Hazelcast, PostgreSQL, RFC 5905 and calendar API documentation.

Activity saga, service workflow and AI authority

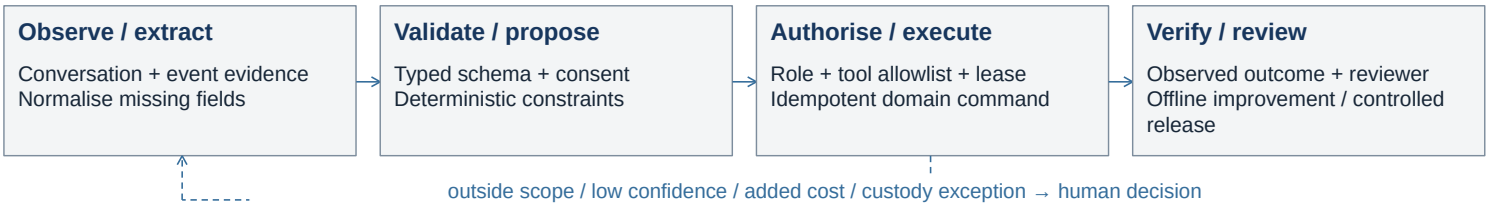
Care outings, after-school care and community events share coordination logic. Each provider retains service responsibility, staffing and handover obligations.

16.1 Swimlane activity flow with parallel holds and partial-failure compensation



Service guards: care = assistance + care-staff capacity; school care = guardian authority + verified handover; community event = venue/stop + host capacity. ETA changes re-enter the saga with a new revision. Provider and transport reservations are not one distributed transaction.

16.2 AI assistance as a constrained tool pipeline



Release stages: shadow comparison → staff-approved actions → bounded automatic tasks
Independent guards enforce consent and accepted terms. Model updates require offline evaluation and versioned release.

Calendar changes produce proposed intent, never auto-booking by default. Sync deletion/resync must not erase D1 bookings. Commercial APIs need access and hold/commit/cancel support; otherwise use an owned manual workflow.

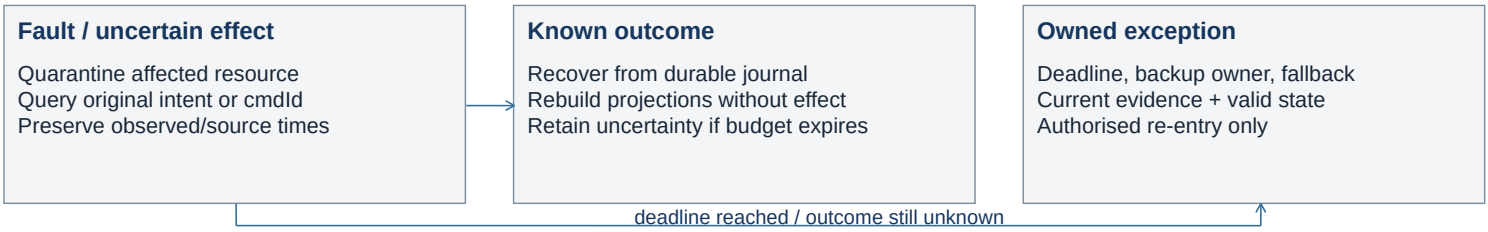
Verification coverage and release dependencies

RM-11. Remote-operation requirements join the PaaS acceptance plan. The cases below are planned tests, not passed results. Release requires reproducible evidence and operating readiness.

17.1 Fault injection and observable remote-operation invariants

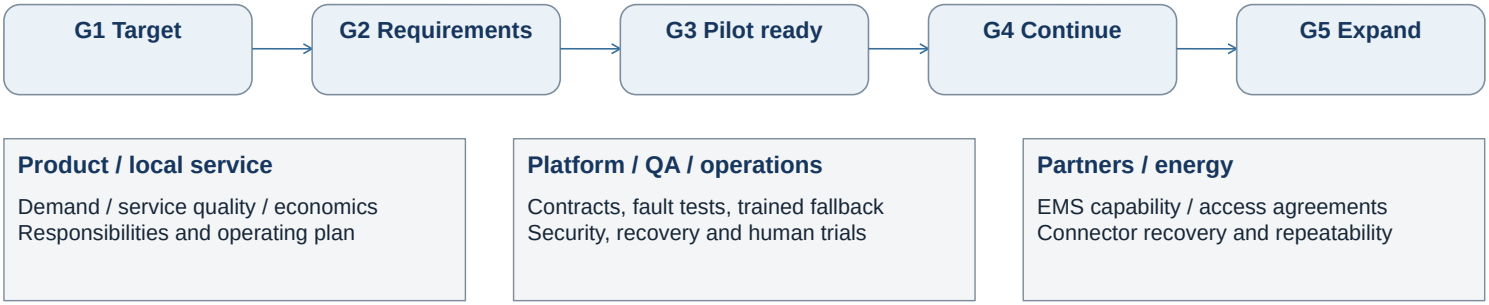
T-R01 / RM-03–04	Delay, reorder, clock jump, reboot	Stale/future data blocks motion; query stays usable.
T-R02 / RM-01,05	ODD limit / sensor degradation	Correct response, no remote extension of ODD.
T-R03 / RM-06,10	Expired / changed-scene advice	Reject with reason and renew context if required.
T-R04 / P13,RM-07	Lose ACK / replay command ID	No duplicate effect; lost result remains reconciled.
T-R05 / RM-02,07	Lose telemetry / media / voice	Independent quality flags; local policy + escalation.
T-R06 / RM-02,10	Centre partition / old operator	Fence ACK + reconcile prior accepted effects.
T-R07 / RM-05,09	SOS + MRM failure / no field team	Emergency and passenger work remain owned.
T-R08 / RM-04,08	Alarm storm / unauthorised AI	No hidden P0; deny AI motion-authorisation.

17.2 Shared PaaS invariants and fault recovery



Add roadside clock/calibration, coverage, false association, reboot and replay tests (P15–P16). P25–P28 extend acceptance to peak demand, spatial imbalance, finite hub capacity, replenishment queues and combined failures. Retain settlement regression tests.

17.3 Evidence chain, milestones and parallel workstreams



G3 evidence: scenario/ODD slice, initial state, revisions, injected fault, trace and oracle result, tested in SIL/HIL/operator/vehicle trials as applicable. G4 uses observed outcomes. G5 adds repeatable onboarding. Record owner, residual risk/restriction and signed decision at each gate.

Service capacity and geographic coverage

Operating feasibility depends on time and location. Fleet and hub totals become usable capacity only when vehicle readiness, travel time and the full service cycle align.

18.1 Demand, reachable supply and commitment

Demand by zone and time

Passenger and shipment OD / windows
Lead time, overlap and peak shape
Party, access, cargo mass / volume
Cancellations and no-shows

→

Feasible supply

Position, ready time and route revision
Seats, payload and cargo space
Passenger detour / arrival limits
Delivery, energy and next-task limits

→

Service commitment

Hold the complete resource bundle
Confirm only after verified commit
Return a binding reason if infeasible
Retain unmet demand in reporting

↓

C03 evaluation at each planning interval

$t_{ready,v} + travelTime(v, origin, departure\ time) \leq latestPickup$
Recompute each existing passenger and delivery deadline after an insertion.
Count shared travel, stop dwell, repositioning, turnaround and replenishment once per vehicle.

18.2 Fleet conservation and workload screen

Ready	Committed	In service	Replenishing	Unavailable
Assignable now	Held / pre-position	Pickup / passenger	Queue / energy	Fault / inspection

$$N_{total} = N_{ready} + N_{committed} + N_{service} + N_{replenishing} + N_{unavailable}$$

Vehicle workload screen

$\Sigma_r\ vehicleHours_r \leq$ available vehicle-hours in the same interval.
Use distinct vehicle routes, including empty travel, dwell and energy stops.
Pooling must not multiply route hours by passenger bookings.
Check segment capacity and local service windows separately.

Spatial and peak-time check

Average capacity does not prove pickup feasibility.
Test demand bursts and directional imbalance by zone.
Hold operational reserve for faults and recovery.
Intercity transfers require explicit travel and authority.

18.3 Hub function, allocation and service outcome

Hub resources

Parking bays and pickup space
Connectors / dispensers / site power
Cleaning and inspection positions
Qualified staff and opening hours

→

Finite allocation

Use time intervals per resource
Sum pooled demand against limits
Reserve only accessible capacity
Propagate queues and outage derating

→

Observable outcome

Passenger / freight fulfillment
Detour, delay and unmet requests
Handling, queue and recovery load
Net contribution / vehicle-hour

Planning record: zone / time interval, OD input, initial vehicle positions, hub resources, availability policy and service windows. Compare feasible alternatives under the same demand trace. A hub count alone does not specify throughput or coverage.

Replenishment and departure readiness

The operating plan reserves replenishment capacity within a vehicle's duty cycle. Daily energy sufficiency and on-time departure are separate acceptance conditions.

19.1 Vehicle cycle with finite replenishment resources

Arrive
Observed energy
Arrival time

→

Queue
Bay / connector
Staff eligibility

→

Replenish
Power or fuel
Actual delivery

→

Turnaround
Clean / inspect
Required evidence

→

Release
Energy + tasks
Next departure

Departure constraint

$t_{arrival} + t_{queue} + t_{replenish} + t_{required_work} \le t_{departure}$
Required work includes passenger / cargo handling, cleaning and inspection as applicable.
Overlap only approved tasks. Recheck route energy and all next departure / delivery commitments.

19.2 Electrical and fuel branches use distinct units

Electrical charging

$E_{depart} = E_{arrive} + \eta_{ch} \int P_{AC}(t)\,dt - E_{aux}$
$E_{next_trip} + E_{reserve} \le E_{depart} \le E_{usable,max}$
P_{AC} in kW, time in hours, energy in kWh.
Use charger / battery limits and charging efficiency.
Energy plan ACK does not prove delivered energy.

Fuel / hydrogen replenishment

$m_{depart} = m_{arrive} + m_{delivered} - m_{consumed}$
$m_{next_trip} + m_{reserve} \le m_{depart} \le m_{usable,max}$
Mass in kg, flow in kg/min, duration in minutes.
Bound usable stock, dispenser flow and turnaround.
Fuel-to-energy conversion needs an explicit model.

SITE LIMITS AT EVERY TIME INTERVAL

$\sum P_{charge,i}(t) \le P_{site,available}(t)$ and occupied connectors ≤ serviceable connectors
$P_{site,available}$ reflects building load, generation, storage and connection limits at the same boundary.
Fuel stock remains non-negative after deliveries and withdrawals. One hub can contain several resources.

19.3 Shortage propagation and recovery

Detect
Power derating / site outage
Low stock / supply delay
Queue growth / delayed arrival
Stale capability or meter data

→

Re-plan
Recalculate departure readiness
Use feasible alternative site / vehicle
Hold affected assignments
Revise ETA and notify service owner

→

Reconcile and release
Observe actual delivered quantity
Complete inspection and required work
Confirm current resource revisions
Release only verified capacity

Acceptance evidence: arrival / departure energy, measured delivery, queue duration, site-limit trace and missed departures. Test concurrent arrivals, one resource unavailable, reduced power and delayed fuel delivery. Report reserve violations and unmet trips even when aggregate daily energy balances.

Degraded operation and recovery capacity

Operational continuity requires bounded service reductions, passenger support and a verified return to service. Recovery uses the same finite fleet, site and staffing resources.

20.1 Failure propagation across the service system

Demand pressure

Peak arrivals / directional imbalance
Longer trips / slow turnaround

Supply loss

Vehicle fault / site outage / energy shortfall
Communication loss / staff absence

↓

Impact assessment

Identify affected bookings and intervals. Recompute reachable vehicles, usable hubs and qualified response capacity.
Preserve already accepted commitments and unknown physical effects. Record unmet demand and passenger exposure.

↓

NORMAL

Service promise feasible
Capacity and evidence current

→

CONSTRAINED

Restrict new offers by zone / time
Protect accepted journeys

→

SUSPENDED

Stop affected new commitments
Maintain support and local fallback

Transitions use configured triggers, persistence and owner approval. Recovery thresholds include hysteresis to avoid repeated entry / exit. Local ADS safety decisions remain independent of this service-mode model.

20.2 Parallel passenger, asset and capacity recovery

Passenger outcome

Contact every affected passenger party
Explain revised ETA / alternatives
Reserve feasible replacement journeys
Track each handover and resolution
Own compensation / refund decision

Vehicle / site outcome

Retain fault and command history
Assign qualified field responder
Observe secure state and clearance
Inspect and record restrictions
Reconcile unresolved physical effects

Resource outcome

Reserve relief / field capacity
Cargo custody / storage / returns
Revise passenger / cargo plans
Include staff travel and queues
Escalate recovery shortfall

20.3 Return-to-service gate and staffing condition

Return-to-service gate

Cause addressed + fresh evidence + authorised release + feasible next mission + passenger outcome resolved.
Connectivity restoration alone cannot release capacity.
Observe sustained recovery before widening offers.

Response capacity

$A = \Sigma \lambda_k E[S_k]$ in matched time units.
$A <$ qualified staffed seats is only a mean-load screen.
Test bursts, common-cause alarms, breaks and travel.
Every open case needs an owner, deadline and backup.
Keep field response separate from remote assistance.

Record trigger, affected scope, admission restriction, accountable owner, next review and recovery evidence for each degraded period.

Operating feasibility and verification evidence

A configuration qualifies only within its tested demand, geography and operating conditions. Service quality, resource limits, recovery and economics require separate evidence.

21.1 Measurement chain and explicit denominators

Requests
Unique service intents

→

Dispatch
Verified assignment

→

Pickup
Physical arrival

→

Completion
Verified trip outcome

Service quality

Dispatch success = assigned / eligible dispatch requests.
Keep capacity and admission rejects in the denominator.
Demand fulfillment = completed / eligible requests.
Report customer cancellations separately by reason.

Pickup and utilization

Wait starts at request time for immediate demand, or agreed pickup time for a reservation. Report p50 / p95 with the observed-arrival count and missing outcomes.
Show on-time pickups / pickups due, including failures.
Separate available hours from scheduled fleet hours.

At cut-off, classify every passenger booking and shipment separately: completed, cancelled, rejected, expired, unresolved or active. A pooled mission is not one passenger request. Retain people counts separately from booking counts. Show exclusions, censoring and missing evidence.

21.2 Reproducible scenario design and pass conditions

Freeze inputs

Passenger / cargo OD and windows
Lead time and sharing acceptance
Party, mass, volume and handling time
Fleet, hubs and response limits
Input / policy / route revisions

→

Exercise conditions

Single-party / pooling / cargo / both
Peak and directional concentration
Vehicle / hub / energy capacity loss
No-show / failed cargo handover
Combined faults and recovery ramp

→

Evaluate each case

Passenger and freight service gates
Seats, payload, volume and energy
Detour, dwell and recovery workload
Unserved / unresolved outcomes
Binding constraint and owner

PASS RECORD

State target, denominator, time window and acceptance rule before running. Use repeatable demand traces or recorded seeds, then independent trials. Preserve every run and failure. Finite synthetic tests establish a tested envelope only. Pilot evidence determines field readiness.

21.3 Capacity decision and business consequence

Feasible alternatives

Compare fleet, hub placement and service policy only after all required operating conditions pass.
Rank cost within the feasible tested set. If none qualifies, report the binding constraint and re-plan.

Economic reconciliation

Reconcile passenger fares and freight revenue.
Count shared route / infrastructure costs once.
Include dwell, displaced trips, handling and returns.
Separate operating contribution, CAPEX and cash flow.

Release evidence: input revision, scenario / seed, execution trace, metric definitions, pass / fail by condition, residual restriction and accountable decision. Operational feasibility does not by itself establish project profitability or safety certification.

Shared rides: route feasibility and passenger outcomes

04. Sharing can improve passenger output per vehicle-hour when compatible journeys fit the same route. Seat occupancy alone does not prove better service or contribution.

22.1 Request compatibility and guarded route insertion

Eligible requests

Pickup / arrival windows
Party size and accessible space
Sharing consent and fare terms
Maximum accepted ride time

→

Candidate insertion

Insert pickup before drop-off
Recompute every downstream ETA
Check capacity on every segment
Keep existing passenger promises

→

Commit and observe

Hold seats and route revision
Confirm accepted terms per party
Observe boarding / alighting
Keep each intent and outcome

22.2 Segment capacity and individual service constraints

Capacity along the route

For each segment s: $\Sigma_i \text{onboard}(i,s) \times \text{partySize}_i \leq \text{seats_usable}(s)$
Check accessible positions and luggage space separately under the vehicle configuration.
A pooled route is one vehicle mission with several passenger bookings. Count its distance once.

Passenger time constraints

$\text{earliestPickup}_i \leq \text{pickup}_i \leq \text{latestPickup}_i$
$\text{arrival}_i \leq \text{latestArrival}_i$
$\text{rideTime}_i - \text{directTime}_i \leq \text{detourAllowance}_i$
Direct time uses the same traffic / time basis.
Include boarding, alighting and intermediate dwell.

Insertion and consent rules

Recheck every onboard and committed passenger.
Protect accepted windows and access requirements.
Offer sharing price and allowed detour explicitly.
An infeasible insertion returns a binding reason.
No-show or cancellation triggers reconciliation.

22.3 Revenue gain, service loss and the comparison basis

Passenger output

Passenger journeys / vehicle-hour
Passenger-km / vehicle-km
Seat-km used / available seat-km
Wait, detour and missed pickups

→

Net contribution

Recognized fares after discounts
Less route, energy and support cost
Include added dwell and recovery
Route cost allocated only once

→

Comparable alternatives

Same underlying passenger intents
Same fleet, hubs and faults
Single-party vs shared operation
Rejected and displaced trips visible

22.4 Failure cases and release evidence

Stress cases

Mismatched OD / low overlap; full accessible space; delayed boarding; no-show after insertion; traffic delay; cancellation while onboard; vehicle loss with several parties needing recovery.
Stop adding riders when an accepted passenger limit or response-capacity constraint would fail.

Evidence: route revision, ordered stops, segment occupancy, accepted detour / fare, actual passenger times and booking-level outcomes. Show total contribution and passenger fulfillment together. Higher occupancy can coexist with fewer completed journeys. Agree thresholds and operating authority before pilot release.

Passenger–freight sharing: capacity and net contribution

05. Cargo uses compatible spare route and vehicle capacity. Incremental freight revenue must cover handling and detour costs while preserving passenger service commitments.

23.1 Cargo eligibility, route insertion and custody

Shipment request

Origin / destination and windows
Mass, volume and compartment needs
Handling time and handover contact
Permitted goods / service policy

→

Joint feasibility

Passenger windows remain binding
Check payload, space and access
Insert loading / unloading dwell
Reserve staff and loading space

→

Verified delivery

Link shipment to route revision
Record pickup / custody handovers
Delivery or failed-attempt proof
Own return, loss and damage cases

23.2 Capacity is a vector at each route segment

Independent constraints

Passenger seats and accessible positions remain within their configured limits.
$\Sigma_j \text{cargoVolume}_{j,s} \le \text{usableCargoVolume}_s$ and $\Sigma_j \text{onboardMass}_{j,s} \le \text{payloadLimit}_s$
Onboard mass includes people, luggage and cargo under the same payload boundary.
Check segregation, securement, door access and passenger comfort before accepting a job.

Time and recovery constraint

Route time = travel + passenger dwell
+ cargo handling + required turnaround.
Recompute every pickup, arrival and delivery ETA.
Cargo waiting / failed handover consumes capacity.
Protect passenger recovery and energy reserve.

Service policy and authority

Use an approved vehicle / goods combination.
Confirm transport, insurance and partner eligibility.
Assign loading, custody and exception owners.
Decline cargo that breaks passenger commitments.
Apply return / storage rules after failed delivery.

23.3 Four operating conditions on the same demand basis

Single-party

No pooling / cargo

Shared rides

Pooling only

Passenger + cargo

Cargo only

Combined

Pooling + cargo

Economic comparison

Δ contribution = Δ passenger revenue + freight revenue − Δ total attributable operating cost
Passenger revenue reflects discounts and displaced / unserved trips. Do not subtract that loss twice.
Include handling, detours, energy, cleaning, support, returns / claims and app cost. Count route cost once.

23.4 Outcome and failure evidence

Report passenger fulfillment / delay and freight delivery / lateness separately, plus vehicle-hours, payload / volume limits, dwell and total contribution. Test full payload, long unloading, absent recipient, incompatible goods, cancellation and vehicle loss. Keep passenger and shipment identities within one shared mission. Hold effects 01–03 constant across the four conditions, then test interactions.